Neuronal Dynamics

Computational study

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Abstract

This study tests whether scalar membrane-potential recordings distinguish regular activity, deterministic chaos and regular activity corrupted by measurement noise. The prescribed Hindmarsh-Rose (HR) system supplies known synthetic sources. Equilibrium continuation, full-state returns and model Lyapunov exponents establish an independent dynamical classification. Scalar reconstructions, finite-scale attractor dimensions and the entropy-complexity plane are then compared with it. A revised chaotic reconstruction gives an effective correlation slope near 1.88. This is a finite-scale estimate: a limiting chaotic D2 and D0 are not established. Independent validation and sampling experiments expose the remaining uncertainty. Ordinal statistics distinguish the selected clean signals, but depend on observation scale and noise. A supplementary Hodgkin-Huxley (HH)/Izhikevich comparison identifies which rates and synchronization properties survive a fixed input calibration. The results support combining complementary methods; they do not establish a universal classifier or an exact fractional dimension from a finite recording.

Problem statement and mathematical formulation

The practical ambiguity is that irregular observations can arise from deterministic instability, a regular source with measurement noise, or nonperiodic motion without chaos. For the prescribed flow $\dot{s} = f(s; I)$, the model-based task is to find all $f(s_*; I) = 0$, their linear spectra, critical currents and attracting regimes. The observation problem is

$$u_n = \pi_x \Phi_I^{T_{tr}+n\Delta t}(s_0), \qquad u_n^{obs} = u_n + \eta\sigma_u\xi_n, \quad \xi_n \sim N(0,1).$$

Parts B-C receive only u_n: hidden coordinates are not supplied to reconstruction or dimension estimators. Delay vectors, correlation sums, box covers and ordinal probabilities are compared with the independent classification in A. The study uses synthetic reference systems and does not assess biological validity or clinical diagnosis. For HH/Izhikevich, a positive affine input transformation is fitted on designated training currents and evaluated on held-out currents.

Experimental design, ablations and statistical units

Numerical verification checks the solution of the equations. Controlled method comparisons identify which analysis component changes an observed result.

Factor	Controlled comparison	Question addressed
Temporal dependence	GP with W=0 versus positive Theiler windows	Bias from nearby times
Temporal ordering	HR recording versus its amplitude-preserving shuffle	Information contained in ordering
Observation scale	Decimation at fixed physical delay versus fixed q	Sampling versus a changed ordinal time scale
Trajectory coverage	Point boxes versus polygonal-path boxes; exact subsamples of one dense trajectory	Cells skipped between observations
Input calibration	Identity mapping versus frozen fitted kappa,beta	Calibration effect on train and valida tion errors
Measurement noise	Clean signal versus nonzero eta, with paired Gaussian draws	Changes in H,C and separation of their means

Integration step, transient, embedding dimension, delay, record length, grid origin and

fitting scales are sensitivity checks, rather than removals of model mechanisms. Computed comparisons are retained in `tables/ablation_sampling_comparison.csv` and `tables/ablation_calibration_scores.csv`. The improved reconstruction was selected exploratorily. Its protocol was then frozen before generating a new IC2 validation recording; validation did not retune it. Independent noise realizations are the resampling units for Monte Carlo intervals. Consecutive time segments provide descriptive variability and are not independent replicates. No p-values are assigned to deterministic grid or convergence checks.

Submission and reproducibility

All required parts A-D and the requested optional section 5 are included. The main submission is this PDF, accompanied by source code, an executed Jupyter Notebook, settings, tables and figures. Markdown files are editable report sources. The assignment permits separate source code and PDF submissions. See README.md for full commands and research.ipynb for the computational narrative. HR variables and time are in model units, not mV and ms; HH/Izhikevich units are specified separately. Stored CSV precision does not imply equivalent inferential accuracy. Every numerical value is obtained from saved calculations.

Equations and numerical protocol

$$\dot{x} = y - x^3 + 3x^2 - z + I, \quad \dot{y} = 1 - 5x^2 - y, \quad \dot{z} = 0.006[4(x + 1.6) - z], \qquad I \in [0, 4].$$

The parameters and the two initial states $(-1.6, -12, 0)$ and $(1, 0, 0)$ are prescribed. The primary solver is classical float64 RK4 with internal step $h = 0.02$. Selected records discard $T_{tr} = 10000$, observe $T_{obs} = 30000$, and save at $\Delta t = 0.2$. Analysis uses $N = 150000$ samples on a half-open interval. Every NPZ retains all state variables, uniform times, events, initial/transient/final states and numerical settings. Discarded transients are reproducible from their initial conditions and duration. Sweep recordings use a coarser output interval of 1; spikes and extrema are still located on the internal integration grid.

Integration and observation steps are varied separately to test numerical and sampling errors. Versions: numpy 1.26.4, scipy 1.16.3, matplotlib 3.10.8, numba 0.62.1, pandas 2.2.3. Complete platform information is in data/settings.json; calculations require no GPU or network.

The original randomized experiments use PCG64 seed 20261024. The additional 30-replicate noise study uses seeds 2026102501 and 2026102502. Initial states, per-realization states and draws are saved in `data/rng_initial_state.json`, `data/noise_rng_states.json` and `data/noise_statistics_settings.json`. Noise and shuffling are applied after solving the ODEs. Test fixtures use independent fixed seeds. Reproduction compares actual numerical arrays and tables.

Numerical accuracy and stationarity

Short integrations on $[0, 20]$ are compared with independent DOP853 at `rtol=2e-13`, `atol=2e-14`:

I	h	max abs error	final l2 error
1	0.04	2.956e-05	2.213e-05
1	0.02	1.635e-06	1.032e-06
1	0.01	1.010e-07	5.376e-08
2.5	0.04	0.000335884	4.266e-05

I	h	max abs error	final l2 error
2.5	0.02	2.767e-05	3.721e-06
2.5	0.01	1.922e-06	2.632e-07
3.25	0.04	0.000692763	0.000114736
3.25	0.02	5.001e-05	8.317e-06
3.25	0.01	3.313e-06	5.521e-07
4	0.04	0.00112697	0.00064511
4	0.02	7.697e-05	4.385e-05
4	0.01	4.985e-06	2.835e-06

Long chaotic trajectories separate pointwise, so convergence is assessed using observables and exponents. Integration steps $0.04, 0.02, 0.01$, transients $5000, 10000, 20000$ and record lengths are checked separately. Full settings and results are in integration_convergence.csv.

regime	h	mean x	std x	mean isi	H	C
spiking	0.04	-0.598908	0.552348	20.1279	0.343952	0.277994
spiking	0.02	-0.59891	0.552349	20.1279	0.343953	0.277995
spiking	0.01	-0.59891	0.552349	20.1279	0.34403	0.278044
bursting	0.04	-1.00609	0.532027	41.2953	0.3533	0.279883
bursting	0.02	-1.00609	0.532026	41.2953	0.353274	0.279879
bursting	0.01	-1.00609	0.532026	41.2953	0.353303	0.279887
chaotic	0.04	-0.796031	0.489837	33.4824	0.394224	0.301216
chaotic	0.02	-0.795555	0.489007	33.562	0.394139	0.301186
chaotic	0.01	-0.796776	0.490754	33.3724	0.394811	0.301555

Reducing $h = 0.02$ to 0.01 changes the regular period by about $4 \cdot 10^{-6}$, chaotic H by about $7 \cdot 10^{-4}$ and the chaotic largest exponent by about $6 \cdot 10^{-5}$ around 0.0135. These differences are small enough for the reported classification, but not for locating every periodic-window boundary precisely. Six consecutive blocks of length 5000 show the finite-record variability:

regime	mean x min	mean x max	std x min	std x max
bursting	-1.00872	-1.00347	0.53069	0.533517
chaotic	-0.798748	-0.79228	0.485038	0.495109
spiking	-0.599316	-0.598238	0.552015	0.552957

Periodic block means differ partly because boundaries contain incomplete cycles. Chaotic variability is retained rather than removed by choosing favorable endpoints.

A. Numerical bifurcation analysis

A1. All equilibria

The stationary equations give

$$y_* = 1 - 5x_*^2, \qquad z_* = 4(x_* + 1.6), \qquad F(x_*; I) = x_*^3 + 2x_*^2 + 4x_* + 5.4 - I = 0.$$

Since $F_x = 3(x + 2/3)^2 + 8/3 > 0$ everywhere and F runs from negative to positive infinity, exactly one real equilibrium exists for every current in $[0, 4]$. Brent's method on the guaranteed bracket $[-3, 1]$ has maximum original-field residual 5.107e-15 on 801 currents. Monotonicity excludes missing equilibrium branches and folds in this range.

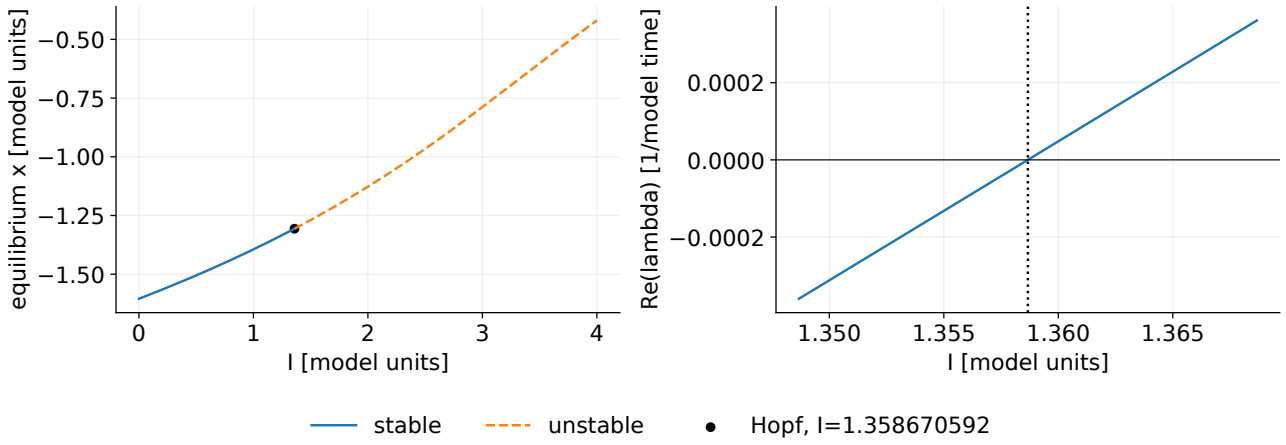


Figure 1: A1-A2: equilibrium branch and stability; solid stable, dashed unstable

Full states, residuals and eigenvalues: equilibria.csv.

A2. Stability and a refined critical point

$$A(x) = \begin{pmatrix} -3x^2 + 6x & 1 & -1 \\ -10x & -1 & 0 \\ 0.024 & 0 & -0.006 \end{pmatrix}.$$

The maximum real eigenvalue crosses zero once along the investigated branch. Brent refinement gives

$$\boxed{I_H = 1.358670591812}, \qquad \omega = 0.0409059652.$$

The opposite-sign bracket is [1.3586705918117, 1.3586705918121], narrower than 10^{-6}. This is numerical localization for fixed coefficients, not a statistical confidence interval. The critical state is -1.306267321, -7.531671566, 1.174930717. The spectrum has one negative real eigenvalue and a simple nonzero imaginary conjugate pair; no zero eigenvalue occurs.

delta I	real	imag
-0.001	-13.9661	0
-0.001	-3.605e-05	0.040906
-0.001	-3.605e-05	-0.040906
-0.0001	-13.963	0
-0.0001	-3.605e-06	0.040906
-0.0001	-3.605e-06	-0.040906
-1.000e-06	-13.9626	0
-1.000e-06	-3.606e-08	0.040906
-1.000e-06	-3.606e-08	-0.040906
0	-13.9626	0
0	1.504e-16	0.040906
0	1.504e-16	-0.040906
1.000e-06	-13.9626	0
1.000e-06	3.606e-08	0.040906
1.000e-06	3.606e-08	-0.040906
0.0001	-13.9623	0
0.0001	3.606e-06	0.040906
0.0001	3.606e-06	-0.040906
0.001	-13.9591	0
0.001	3.606e-05	0.0409059
0.001	3.606e-05	-0.0409059

The transverse crossing derivative is 0.03605521, using a centered current step 10^{-5}. With $\|q\|_2 = 1$, $p^H q = 1$, the first Hopf normal-form coefficient is $l_1 = 0.01387674 > 0$, calculated as

$$l_1 = \frac{1}{2\omega}\Re\langle p, C(q,q,\bar{q}) - 2B(q, A^{-1}B(q,\bar{q})) + B(\bar{q}, (2i\omega\mathbf{1} - A)^{-1}B(q,q))\rangle,$$

where $B(u,v) = ((-6x_* + 6)u_x v_x, -10u_x v_x, 0)^T$ and $C(u,v,w) = (-6u_x v_x w_x, 0, 0)^T$. This coefficient is not an attractor Lyapunov exponent. These checks support a nondegenerate **subcritical Hopf bifurcation**: the equilibrium loses stability as I increases and a small unstable cycle is expected on its stable side. The unstable cycle and a cycle-fold location were not continued in this study.

A3. Maxima, interspike intervals and four sweep protocols

The grid $I = 0, 0.05, \ldots, 4$ has 81 points. Every observation lasts 4000. Transients are 6000 except at $I = 1.30$ (12000) and $I = 1.35$ (80000), where slow decay invalidated the original short protocol. Forward and backward sweeps propagate the actual final state; each independent run restarts from one prescribed IC. Spikes are positive crossings of $x = 0$, with within-step cubic Hermite localization. Local maxima are positive-to-negative crossings of $\dot{x}$, including subthreshold extrema; they are not selected by a membrane-potential threshold.

All 109 quiet tails among 324 records satisfy full-state distance below $1.2 \cdot 10^{-10}$ and field residual below $4.8 \cdot 10^{-12}$. Raw extrema in numerically flat tails (x range $\leq 10^{-9}$) are retained but are not drawn as an oscillatory branch. Rest is represented by A1's equilibrium branch. Per-record diagnostics: sweep_stationarity.csv.

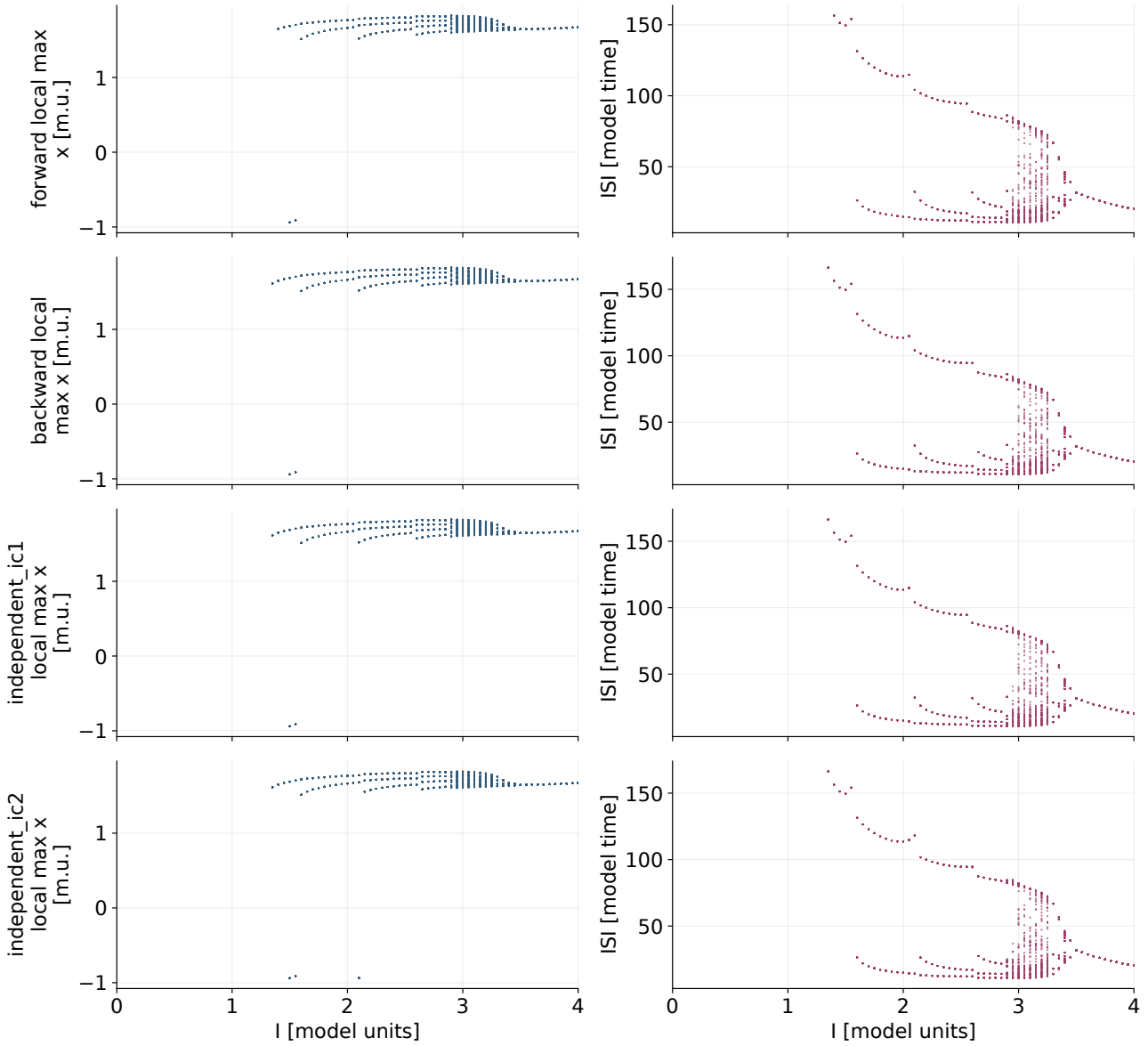


Figure 2: A3: four protocols; local maxima on the left and ISI on the right

protocol	largest I without spikes	smallest I with spikes
backward	1.3	1.35
forward	1.35	1.4
independent_ic1	1.3	1.35
independent_ic2	1.3	1.35

The forward sweep remains quiet at 1.35 and spikes at 1.4; the other protocols already spike at 1.35. These are width-0.05 grid brackets, not refined bifurcation values. A refined region $I \in [2.8, 3.4]$ uses 31 points at step 0.02, transient 10000 and observation 12000; it reveals irregular bands and periodic windows.

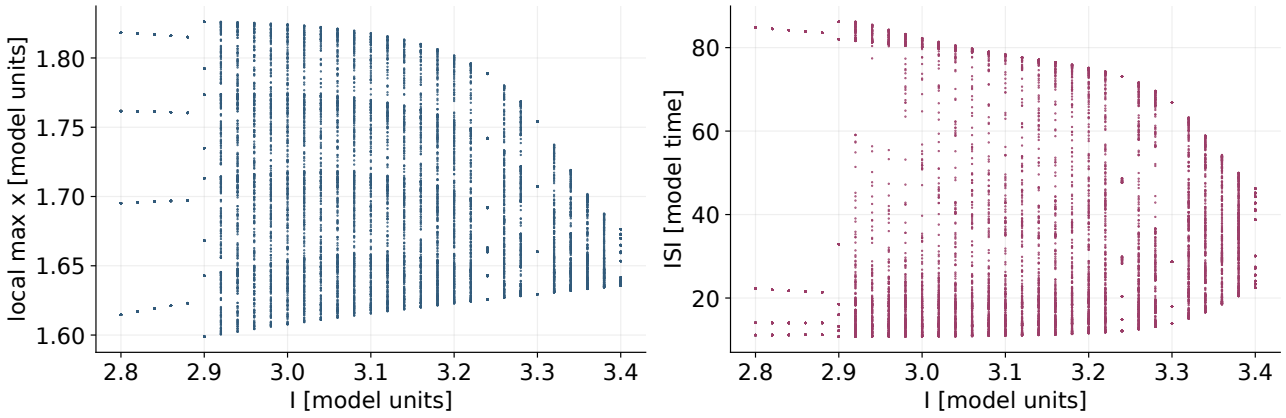


Figure 3: A3: refined regime-transition region

Near onset, separate 40000-transient runs include the prescribed ICs and small perturbations of the known equilibrium:

I	initial	spikes	range x	distance to eq
1.33	ic1	34	3.23837	0.146804
1.33	ic2	34	3.23977	1.05603
1.33	near_eq	0	0	3.140e-12
1.358	ic1	36	3.25005	0.719986
1.358	ic2	36	3.25003	1.35843
1.358	near_eq	0	5.500e-07	2.992e-06
1.359	ic1	37	3.24939	2.06322
1.359	ic2	37	3.24983	3.62316
1.359	near_eq	0	2.499e-06	1.570e-05
1.37	ic1	37	3.25062	3.05541
1.37	ic2	37	3.25087	4.17585
1.37	near_eq	37	3.25213	1.52729

At $I = 1.33 < I_H$, the prescribed ICs oscillate while a state near the equilibrium converges to rest, demonstrating coexistence. Small amplitudes after 40000 near Hopf do not establish stability; two perturbation runs were extended to 10^6:

I	transient	spikes	range x	distance to eq
1.358	1000000	0	9.822e-12	6.248e-11
1.359	1000000	37	3.24985	1.80474

Perturbations decay below Hopf and escape above it. Activity onset in a sweep thus need not coincide with equilibrium destabilization. See onset_statistics.csv and onset_extended.csv.

A4. Independent dynamical classification

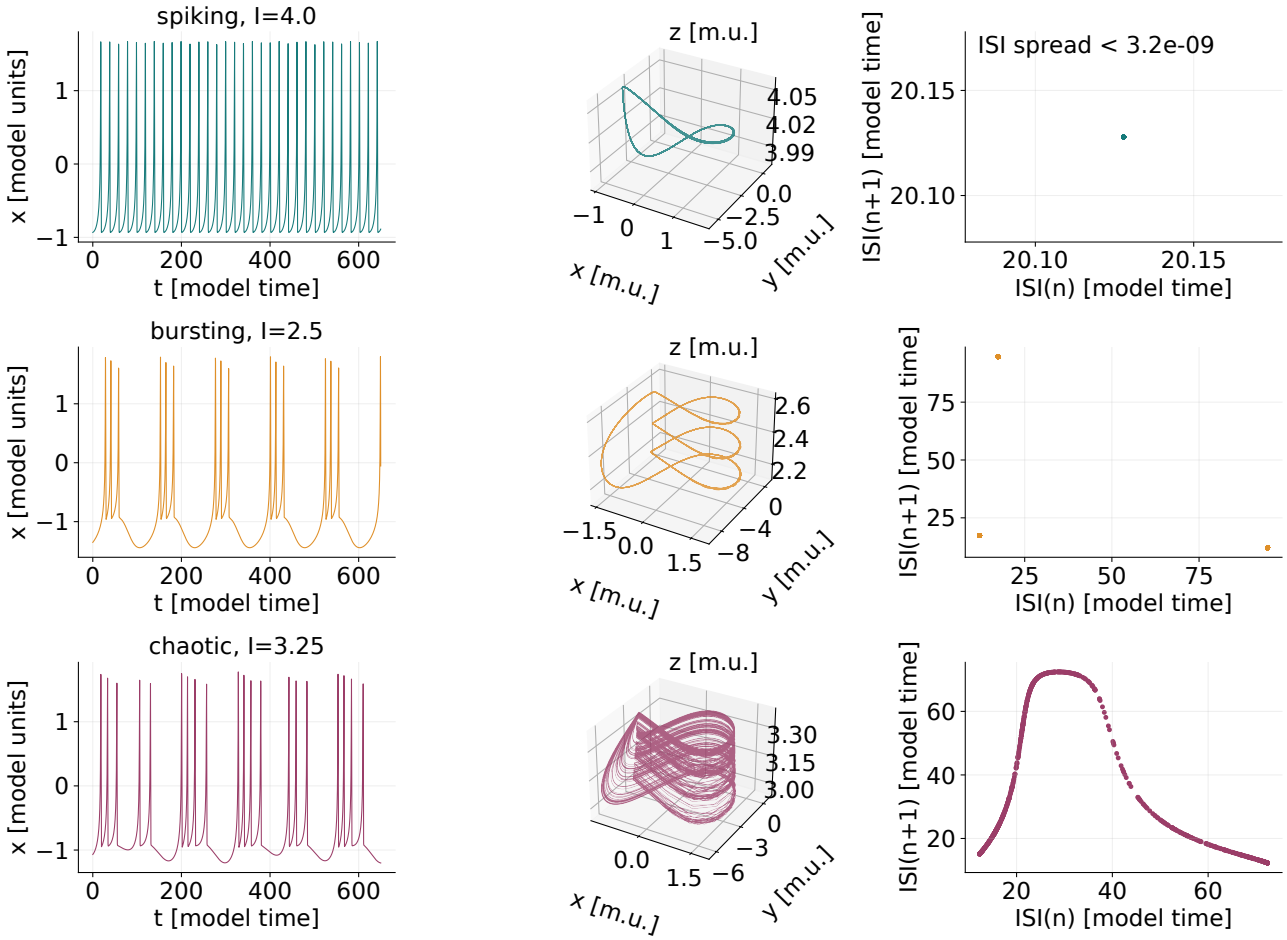


Figure 4: A4: time series, full-state phase portraits and ISI return plots

The regular spiking section returns to one full-state point; bursting returns to three, with a full period equal to three ISIs. All three state components are checked. No shift by 1-12 spikes closes the chaotic candidate comparably well.

regime	p spikes	mean period	std period	section median error	section max error
spiking	1	20.1279	1.102e-09	2.092e-09	3.043e-09
bursting	3	124.106	1.183e-09	2.213e-09	4.866e-09
chaotic	4	134.411	19.0913	0.0636762	0.374202

The variational equation $\dot{Q} = A(s(t))Q$ is integrated jointly with the state by RK4. Modified Gram-Schmidt orthogonalization occurs every unit of model time. After 10000 warm-up, accumulated spectra are observed for 80000, with saved 1000-time blocks. Steps $0.04, 0.02, 0.01$ and chaotic QR intervals $0.5, 1, 2$ are compared.

regime	h	lambda1	lambda2	lambda3	first half mean	second half mean	sum minus divergence
spiking	0.04	3.390e-05	-0.0418228	-6.54761	4.126e-05	2.654e-05	0.000978763
spiking	0.02	3.436e-05	-0.0418219	-6.54855	4.166e-05	2.706e-05	5.283e-05
spiking	0.01	3.438e-05	-0.0418218	-6.5486	4.168e-05	2.709e-05	3.070e-06
bursting	0.04	4.837e-06	-0.00463916	-10.9137	4.290e-05	-3.323e-05	0.0127878
bursting	0.02	4.877e-06	-0.00463909	-10.9259	4.308e-05	-3.333e-05	0.000625725
bursting	0.01	4.880e-06	-0.00463908	-10.9265	4.309e-05	-3.333e-05	3.463e-05
chaotic	0.04	0.0131977	2.646e-05	-8.41381	0.0132244	0.013171	0.00275643

regime	h	lambda1	lambda2	lambda3	first half mean	second half mean	sum minus divergence
chaotic	0.02	0.013477	-5.766e-06	-8.40903	0.0132273	0.0137268	0.000142809
chaotic	0.01	0.0135346	-3.593e-06	-8.40861	0.0134078	0.0136614	8.146e-06

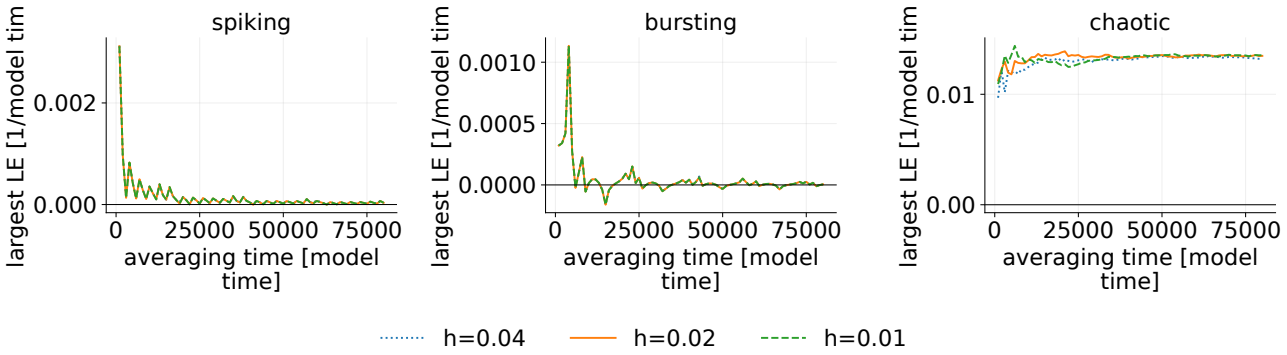


Figure 5: A4: largest-exponent convergence

An autonomous limit cycle has a neutral direction; tiny positive finite-time estimates for periodic signals do not imply chaos. At $I = 3.25$, the largest exponent remains positive, the second is near zero and the third negative. Both half-record averages are positive at every step. Together with bounded irregular motion and a nonperiodic section, this supports deterministic chaos numerically. The sum is checked against mean divergence (`sum_minus_divergence`); the most negative exponent is less accurate at $h = 0.04$. Sign conclusions use the finer steps. For rest at $I = 1$: $\|s_{end} - s_*\|_2 = 6.71e-13$, $\max x - \min x = 0$, $\|f(s_{end})\|_\infty = 4.44e-14$.

Independent solver and periodic-orbit verification

A separately written vector field/Jacobian is checked by complex-step differentiation (maximum error 1.78e-15). An independent 80-digit Decimal Routh-Hurwitz calculation $a_1 a_2 = a_3$ gives $I_H = 1.358670591811940834\ldots$, agreeing with A2. This arithmetic precision does not represent uncertainty in biological parameters. Full-state shooting on $x = 0$ refines periodic orbits; independent DOP853 verifies them:

regime	period	closure residual	neutral multiplier error	transverse multiplier
spiking	20.1279	2.245e-12	1.162e-11	0.431018
bursting	124.106	5.987e-13	5.274e-13	-0.562256

The neutral multiplier is near 1 and the transverse multiplier has modulus below 1. Strong contraction is ill-conditioned in a raw monodromy eigendecomposition, so its logarithm is saved using Liouville's formula. Perturbing the period by $\pm 10^{-5}$ increases closure residuals to $7\ldots9 \cdot 10^{-6}$. Independent DOP853 plus Householder QR gives chaotic $\lambda_1 = 0.012496$ over 5000 after warm-up, with all five 1000-time blocks positive (0.00820-0.01688). It verifies sign and scale, rather than exact agreement with another trajectory's 80000-time mean. See independent_hr_checks.csv.

Library reference calculations

Standard calculations are also performed with established research packages. The ordpy implementation computes ordinal distributions, normalized permutation entropy and Jensen-Shannon complexity from the actual scalar recordings. Exact lexicographic

ranks implement the stated (value, time-index) tie rule without random jitter. This includes rounded ISIs. Across 66 signal/settings combinations, every full-alphabet count agrees; the maximum absolute H,C difference is 1.17e-15. The notebook computes its principal H,C comparison directly through ordpy. See `tables/library_ordinal_validation.csv`.

JiTCODE forms the HR tangent equations by symbolic automatic differentiation and integrates them with SciPy's adaptive DOP853 backend. Its own QR implementation estimates all three model exponents. The initial tangent basis is the identity, with transient 10000, observation 30000 and QR every one model-time unit. Two tolerance settings are compared below. All state coordinates, local growth rates and tangent bases are retained.

regime	rtol	lambda1	lambda2	lambda3	block min	block max
spiking	1.000e-08	7.734e-05	-0.0418381	-6.54903	-0.000499189	0.00042159
spiking	1.000e-10	7.734e-05	-0.0418381	-6.54903	-0.000499189	0.00042159
bursting	1.000e-08	1.253e-05	-0.00462463	-10.9237	-0.000513711	0.000684805
bursting	1.000e-10	1.253e-05	-0.00462463	-10.9237	-0.000513711	0.000684805
chaotic	1.000e-08	0.0133477	5.253e-05	-8.41465	0.0116681	0.0149931
chaotic	1.000e-10	0.0141381	-2.040e-05	-8.40323	0.0129453	0.0151751

The periodic estimates remain compatible with zero. Both chaotic runs have positive means and positive 5000-unit block means; their different finite-time values reflect different chaotic trajectories and do not establish accuracy in the last displayed digits. Block ranges are descriptive, not independent-sample confidence intervals. This verifies the sign and scale independently of the fixed-step code; it does not strengthen the uncertain limiting D0/D2 claims.

SciPy already supplies adaptive ODE solvers, root finding, spatial trees and linear algebra throughout the study. Explicit custom layers enforce the assignment's Theiler windows, scaling diagnostics, and convergence experiments. The hybrid Izhikevich tangent calculation additionally requires event saltation, which a generic smooth-flow exponent routine cannot replace. Exact package versions: ordpy 1.2.2, jitcode 1.7.3, jitcxde_common 1.5.6, symengine 0.14.1, scipy 1.16.3.

B. Scalar reconstruction and attractor dimensions

B1. Preprocessing, ACF, mutual information and false neighbors

Only post-transient x_n enters B. A single mean and standard deviation standardize each series; no signal smoothing, hidden coordinates or phase fitting is used. Fixed nonsingular affine scaling preserves theoretical dimensions.

$$X_i^{(m)} = (x_i, x_{i+k}, \dots, x_{i+(m-1)k}), \quad 0 \le i < N - (m-1)k.$$

The ACF uses FFT and a common zero-lag normalization; its candidate is the first nonpositive lag. Mutual information uses 16/32/64 equal-width histogram bins and marginals from the same joint histogram. Only the diagnostic MI curve is smoothed (Gaussian sigma 1 sample, prominence 0.005 nat) to identify a pronounced first minimum; raw curves are retained. ACF decorrelation and MI dependence need not agree.

regime	ACF lag	first MI lag	initial selected lag	tau	initial m	W samples	FNN
spiking	16	37	37	7.4	3	148	0.00491667
bursting	159	54	125	25	5	500	0.00266667

regime	ACF lag	first MI lag	initial selected lag	tau	initial m	W samples	FNN
chaotic	124	164	164	32.8	8	656	0.0883333

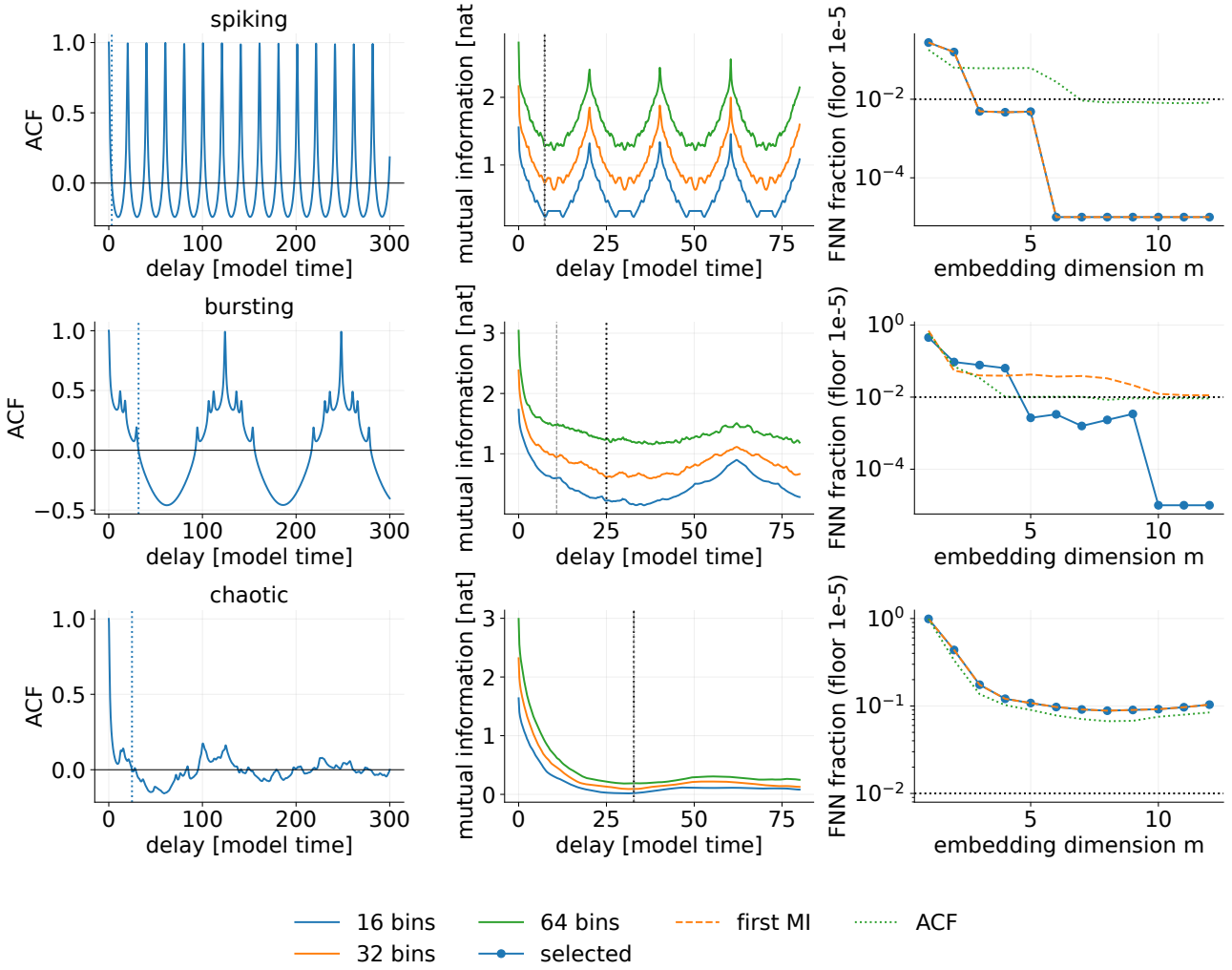


Figure 6: B1: original ACF, MI and FNN diagnostics

FNN uses the same eligible start indices for m and m+1 and the same nearest neighbor. Neighbors with $|i-j| \leq W$ are excluded. A neighbor is false if $|x_{i+mk} - x_{j+mk}|/r_m > 10$ or $r_{m+1}/\sigma_x > 2$. Zero distances are handled explicitly. Up to 12000 indices cover the whole record; W remains in original sample units. The operational criterion is below 1% FNN, sustained at higher m. The bursting first-MI choice fails; its third MI32 minimum $k = 125$ produces sustained FNN below 1% from m=5. Initial candidates remain visible.

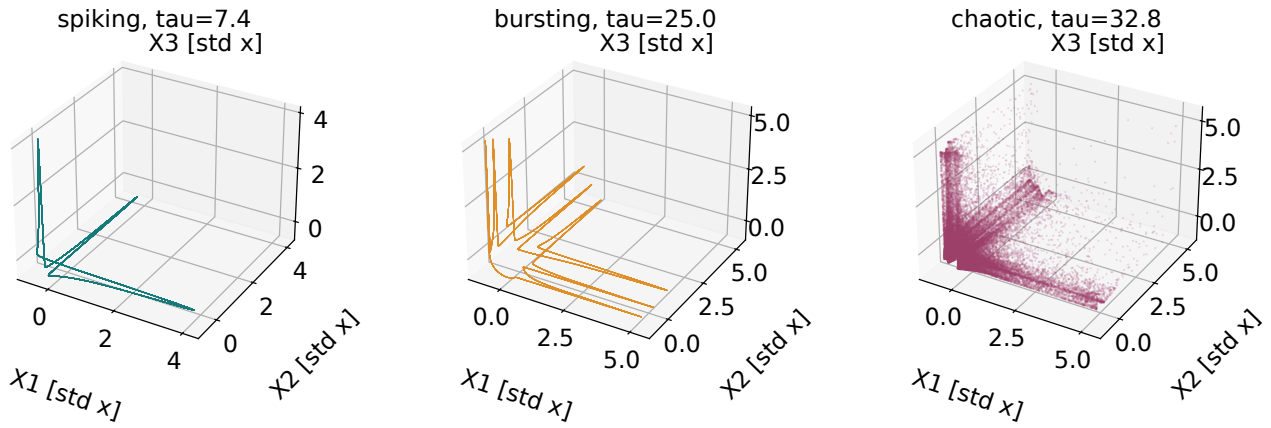


Figure 7: B1: initial delay projections, including the unsuccessful chaotic first-MI choice

For chaos, $k = 164$ ($\tau = 32.8$) still gives about 8.8% FNN at trial m=8; that initial embedding is unconfirmed. Shorter candidates k=8,10,12, tested with the same criteria and W=656, support a conservative common m=6. At k=10, adjacent correlation is about 0.355 and covariance condition number about 3.73, excluding a simple near-diagonal collapse as the explanation of improved FNN.

lag	tau	m	W	FNN	usable
8	1.6	6	656	0	12000
10	2	6	656	0.0005	12000
12	2.4	6	656	0.00775	12000
124	24.8	6	656	0.07775	12000
164	32.8	6	656	0.0973333	12000

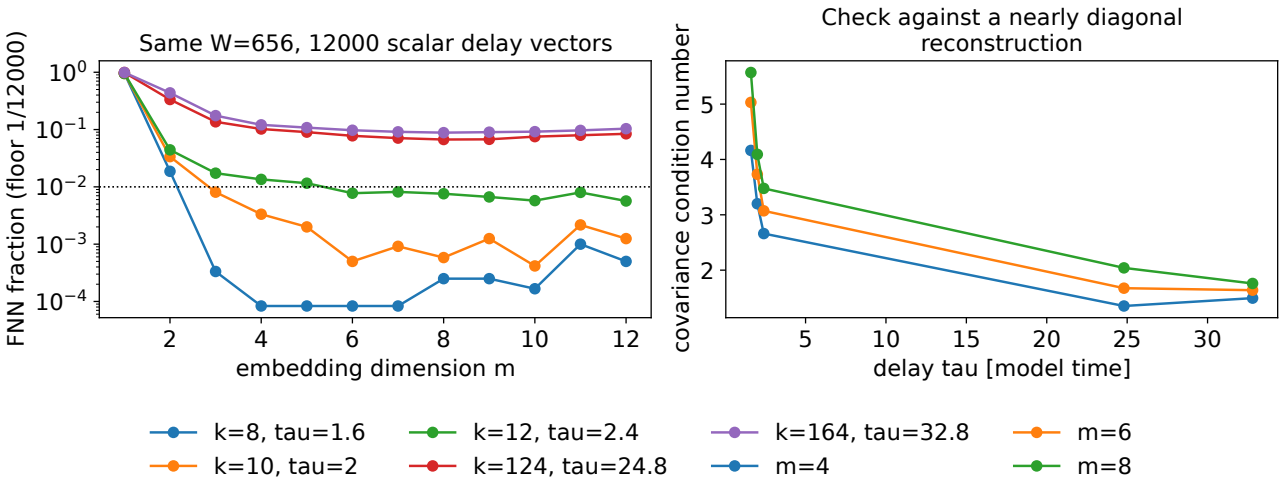


Figure 8: B1: shorter delays, FNN and covariance conditioning

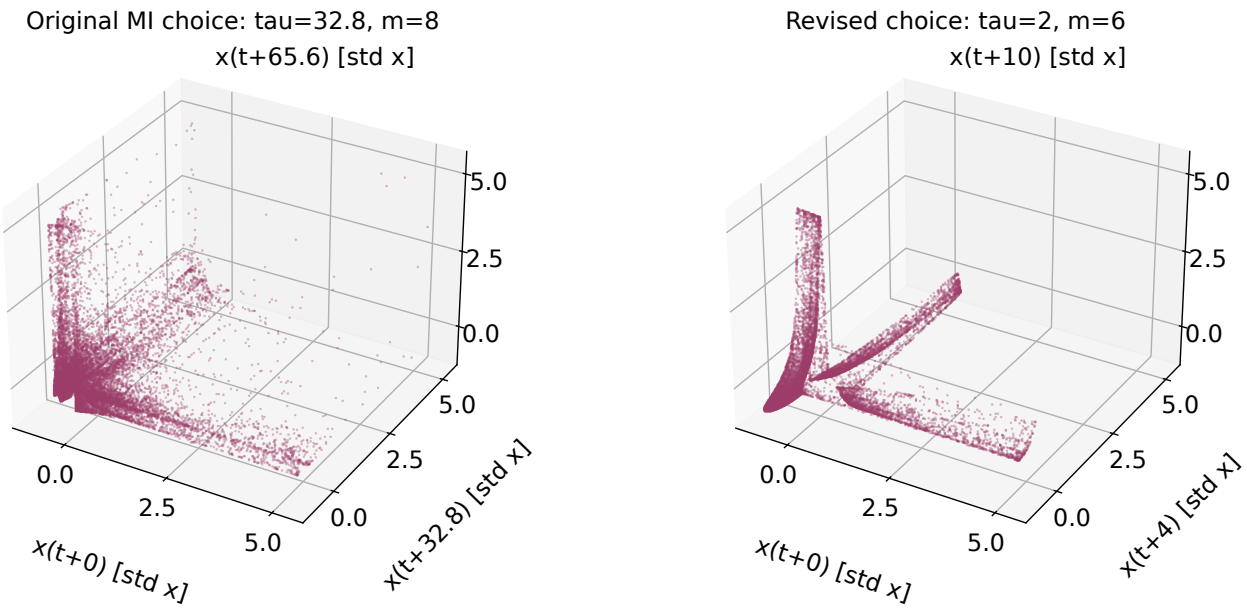


Figure 9: B1: initial MI and revised chaotic delay geometry

The revised chaotic analyses use k=10,m=6,W=656 (physical delay 2, exclusion time 131.2). The original justified regular-signal choices remain. These diagnostics support unfolding but do not prove an embedding theorem for the finite recording. Settings: settings.json.

B2. Correlation dimension and temporal dependence

For an invariant measure, $D_2 = \lim_{r\downarrow 0} \ln C(r)/\ln r$ when the limit exists. The Grassberger-Procaccia estimate uses Euclidean distances and an explicit Theiler exclusion:

$$\widehat{C}_m(r;W) = \frac{\sum_{a<b,\ i_b-i_a>W} \mathbf{1}\{\|X_{i_a} - X_{i_b}\| \le r\}}{\#\{a<b : i_b - i_a > W\}}.$$

`cKDTree.count_neighbors` is converted to unordered pairs, with diagonal and temporal pairs removed from numerator and denominator. This exact sum over selected vectors is checked against brute force. Original calculations use up to 12000 vectors; refined calculations use up to 16000. Radius is shown as $r/\sqrt{m}$ in standardized scalar units. Fits require at least 200 close pairs and $C < 0.15$.

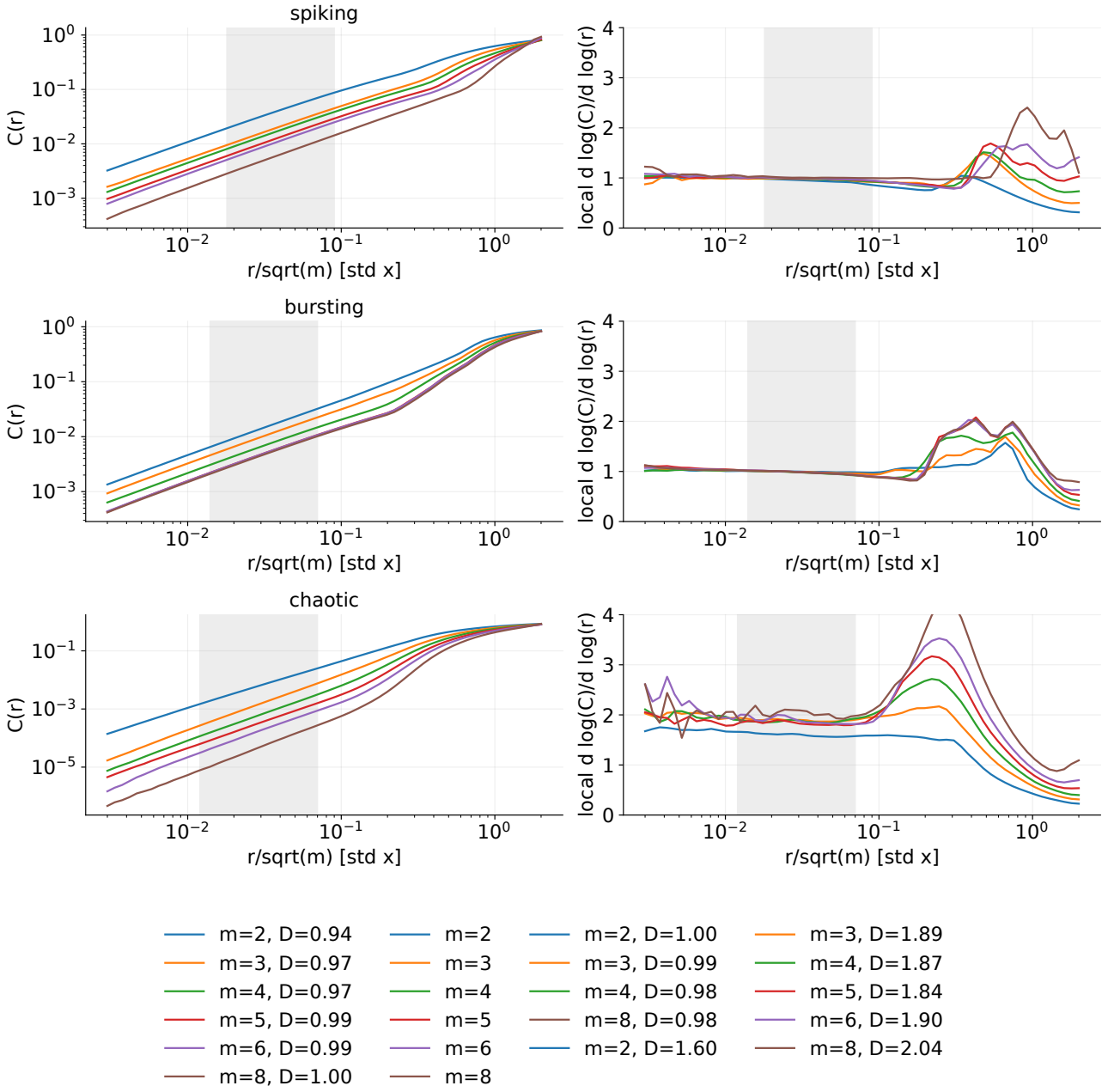


Figure 10: B2: original correlation sums and local slopes; chaotic first-MI results are diagnostic only

regime	m	estimate	local min	local max
spiking	2	0.939333	0.873643	0.979332
spiking	3	0.970337	0.923751	0.988296
spiking	4	0.974586	0.933276	1.00038
spiking	5	0.985604	0.950915	1.00728
spiking	6	0.988077	0.957325	1.00409
spiking	8	1.00437	0.996952	1.02322
bursting	2	0.995961	0.982662	1.01008
bursting	3	0.988968	0.959243	1.01152
bursting	4	0.980172	0.926316	1.01233
bursting	5	0.986816	0.936771	1.01983
bursting	6	0.985718	0.93493	1.02137
bursting	8	0.983727	0.93191	1.01454

The revised reconstruction compares neighboring delays, m=3,4,6,8, W=0,328,656,

1312,2624, partial and doubled records, and neighboring radius intervals. The main interval is $0.001 \le r/\sqrt{m} \le 0.01$; the k=10,m=6 fit is 1.8767, local slopes 1.825-1.974, minimum 2437 close pairs.

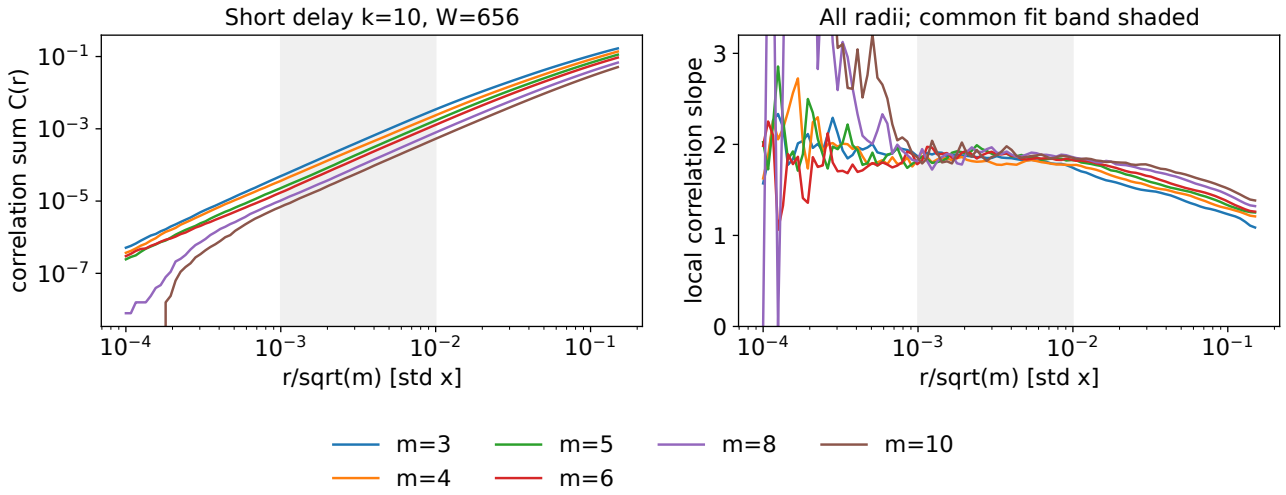


Figure 11: B2: refined correlation sums, local slopes and fitting bands

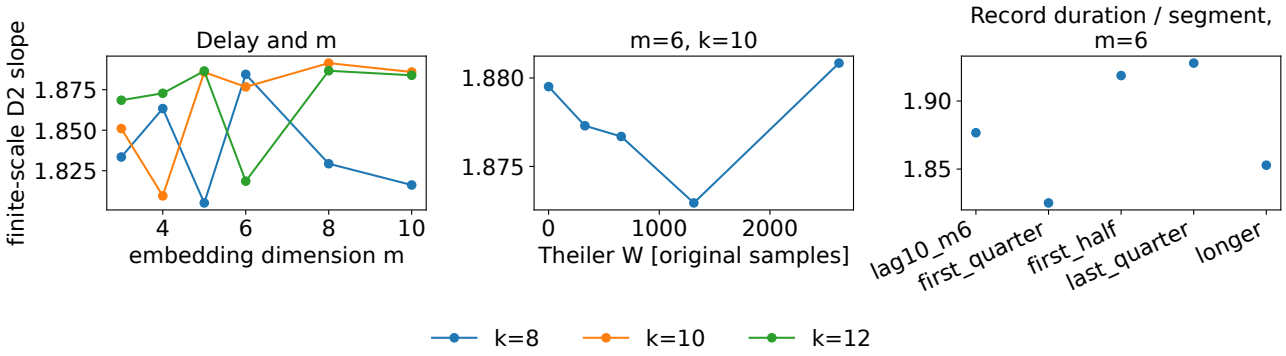


Figure 12: B2: sensitivity to delay, W, record length and radius band

varied	estimate min	estimate max	local min	local max	comparisons
N	1.82511	1.92789	1.74197	2.10904	4
W	1.87294	1.88084	1.823	1.98952	4
lag at m=6	1.81855	1.88444	1.76113	1.97353	3

These are sensitivity envelopes, not confidence intervals. Curves across radii and embeddings reuse the same observations; high regression R-squared cannot establish an asymptotic dimension. The results show geometry more complex than a smooth cycle and an effective slope around 1.9. An exact limiting D2 is not established. All 130 fits: reconstruction_refinement_gp.csv.

B3. Capacity dimension and incomplete coverage

Capacity dimension is $D_0 = \lim_{\varepsilon \downarrow 0} \ln N(\varepsilon) / \ln(1/\varepsilon)$ when the limit exists. Point box-counting counts distinct cells $\lfloor (X - b)/\varepsilon \rfloor$. The original study compares m=3,4,5, lengths N/4,N/2,N and four grid shifts 0,0.25,0.5,0.75 cell widths, using the same scalar normalization in every coordinate. The regular spiking slope is consistent with a curve:

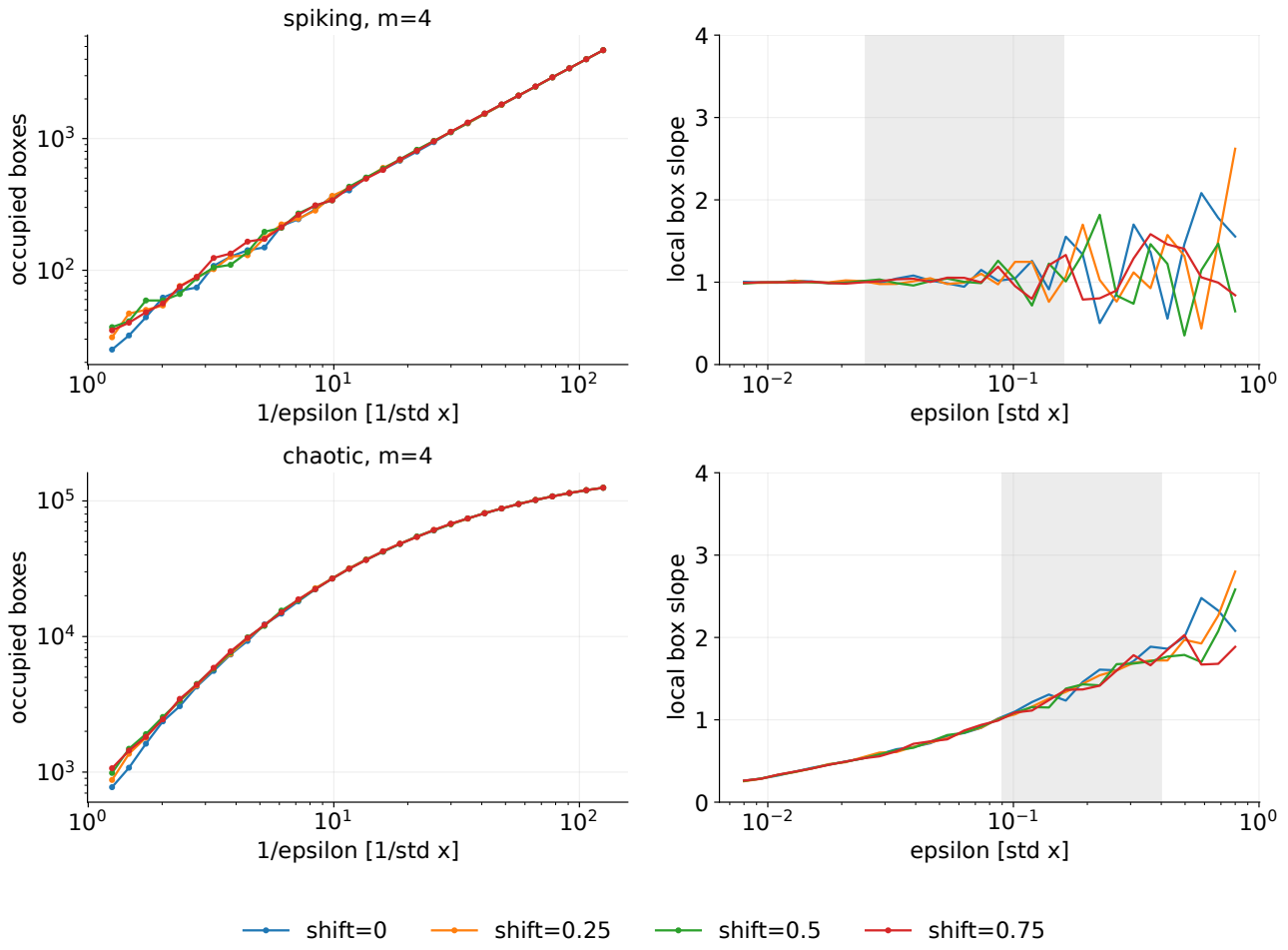


Figure 13: B3: original point box-counting and fitting bands

m	record N	min	max
3	37500	1.04477	1.09469
3	75000	1.04486	1.09411
3	150000	1.04564	1.09544
4	37500	1.00776	1.04606
4	75000	1.01582	1.04824
4	150000	1.01701	1.05046
5	37500	0.996728	1.01975
5	75000	1.00809	1.02392
5	150000	1.00998	1.02683

For chaos, original first-MI m=4 point fits give 1.410-1.452, below the correlation estimate. The inequality $D_2 \leq D_0$ concerns limiting dimensions; its violation by finite fits signals inadequate coverage or scales. The failed attempt is retained, and intervals are not tuned to force agreement. A separate estimator counts all cells crossed by segments joining consecutive scalar delay vectors. This covers a polygonal path, adds no independent observations and uses no hidden states. Exact traversal is tested on analytic lines, boundaries and mixed-direction corners. Every finite polyline has limiting dimension 1; only its resolved intermediate-scale behavior is examined here.

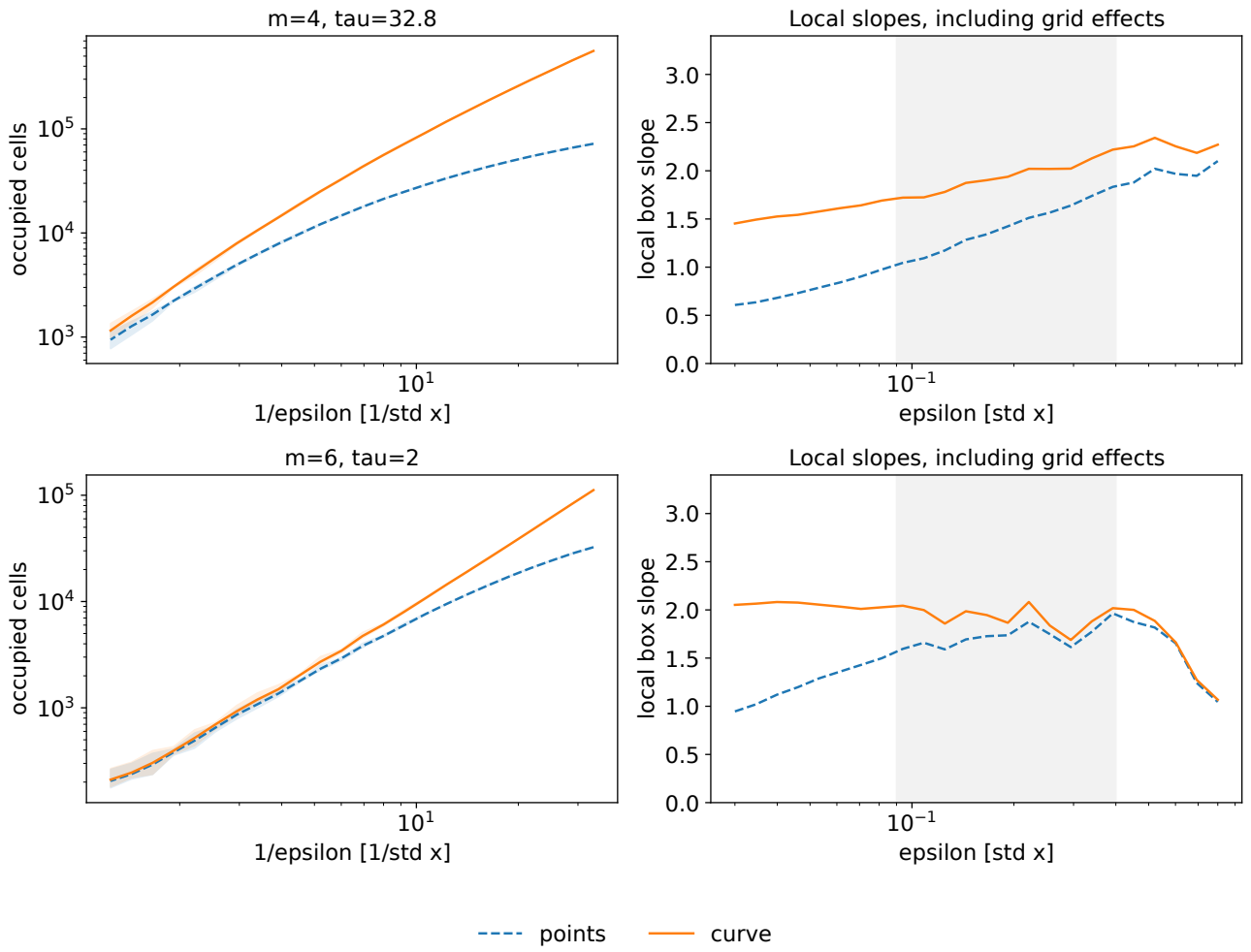


Figure 14: B3: points versus polygonal paths, record length, embedding and grid origin

For k=10,m=6 on $0.09 \leq \varepsilon \leq 0.4$:

case	dt	method	estimate min	estimate max
main	0.2	curve	1.87433	1.94097
main	0.2	points	1.68877	1.74657
main_finer	0.04	curve	1.84286	1.90542
main_finer	0.04	points	1.7931	1.86204
main_longer	0.2	curve	1.88183	1.94899
main_longer	0.2	points	1.74961	1.80732

A single dense trajectory uses h=0.0025, observation step 0.005, transient 10000 and observation 10000. All states are retained; only x is analyzed. Exact subsamples at 0.01,0.04,0.2 isolate observation density from changes in the chaotic orbit, while preserving physical delay and normalization. At m=6,tau=2:

dt	method	estimate min	estimate max
0.005	curve	1.87792	1.88917
0.005	points	1.86419	1.8715
0.01	curve	1.87775	1.889
0.01	points	1.85364	1.86528
0.04	curve	1.87925	1.88877
0.04	points	1.78494	1.79602
0.2	curve	1.90392	1.91485
0.2	points	1.54546	1.57503

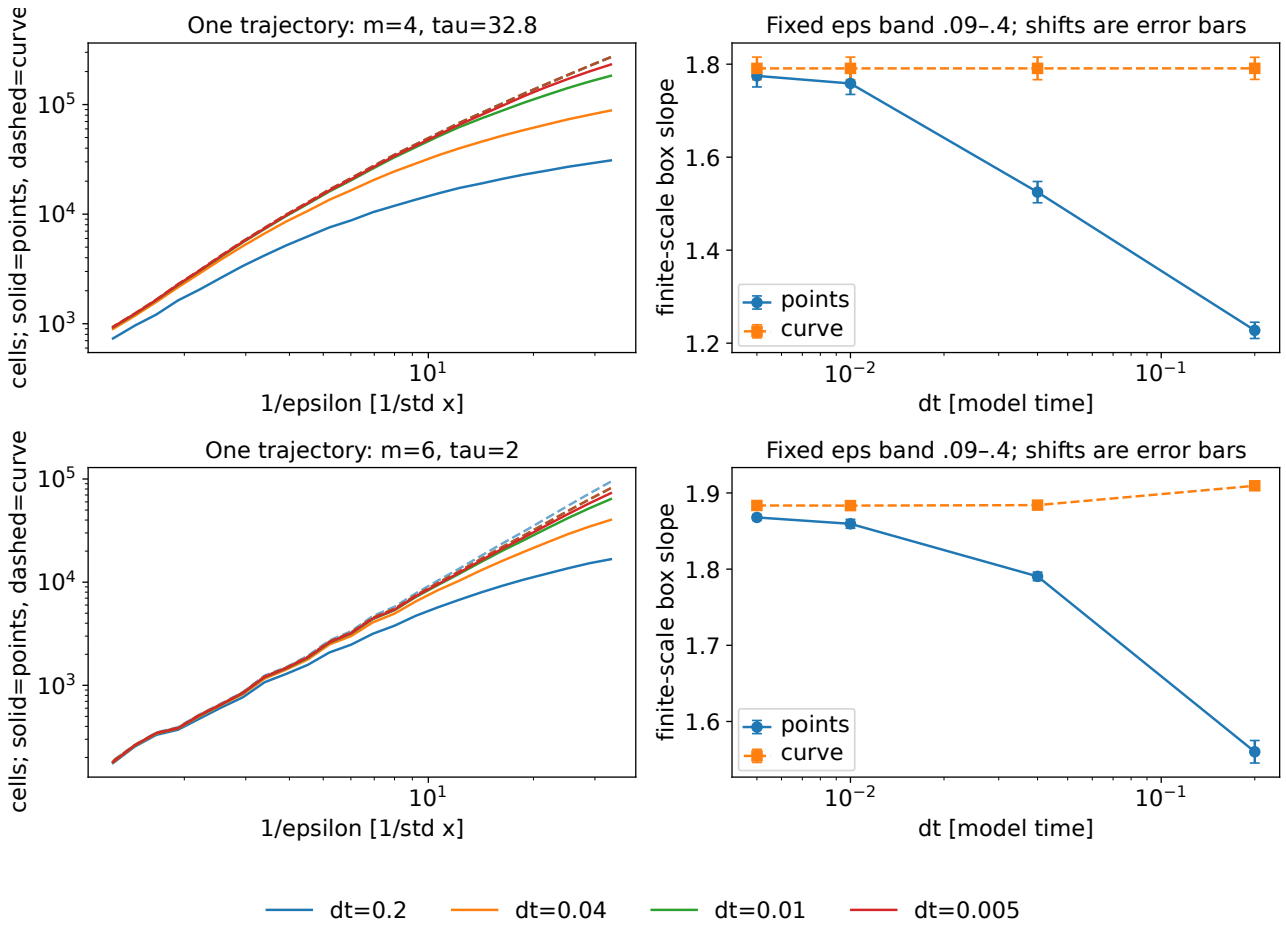


Figure 15: B3: one dense trajectory and its exact subsamples

Point estimates approach path estimates as observation becomes denser. Path fits at steps 0.005 and 0.01 nearly agree; step 0.2 has a shift around 0.025. The maximum dense delay-path step is about 0.052 scalar standard deviations, below the minimum fitted cell width 0.09. This checks sampling resolution; finite duration and transverse coverage remain limitations. Wide local-slope variation prevents interpreting the mean slope near 1.9 as an exact limiting capacity dimension.

B4. Frozen-protocol validation and interpretation

After exploratory selection, k=10,m=6,W=656, GP band [0.001,0.01] and box band [0.09,0.4] were frozen, with a source snapshot and SHA256 hashes. A new recording was then generated from the second prescribed state (1,0,0), at I=3.25, transient 10000, observation 30000, dt=0.2 and h=0.02. Normalization remains fixed from IC1. No settings changed after IC2 was observed. Training sensitivity envelopes are not probabilistic prediction intervals; out-of-envelope results remain disclosed.

metric	setting	validation min	validation max	disclosed lower	disclosed upper	inside disclosed envelope
FNN	k=8,m=6	0	0	0	0.01	True
FNN	k=10,m=6	0.0005	0.0005	0	0.01	True
FNN	k=12,m=6	0.00958333	0.00958333	0	0.01	True
D2_effective	k=10,m=4	1.87208	1.87208	1.80949	1.87275	True
D2_effective	k=10,m=6	1.90959	1.90959	1.81855	1.92789	True
D2_effective	k=10,m=8	1.95677	1.95677	1.82932	1.89137	False
D0_effective	points, k=10,m=6	1.67448	1.73161	1.68877	1.86204	False

metric	setting	validation min	validation max	disclosed lower	disclosed upper	inside disclosed envelope
D0_effective	curve,k=10, m=6	1.87359	1.9358	1.84286	1.94899	True

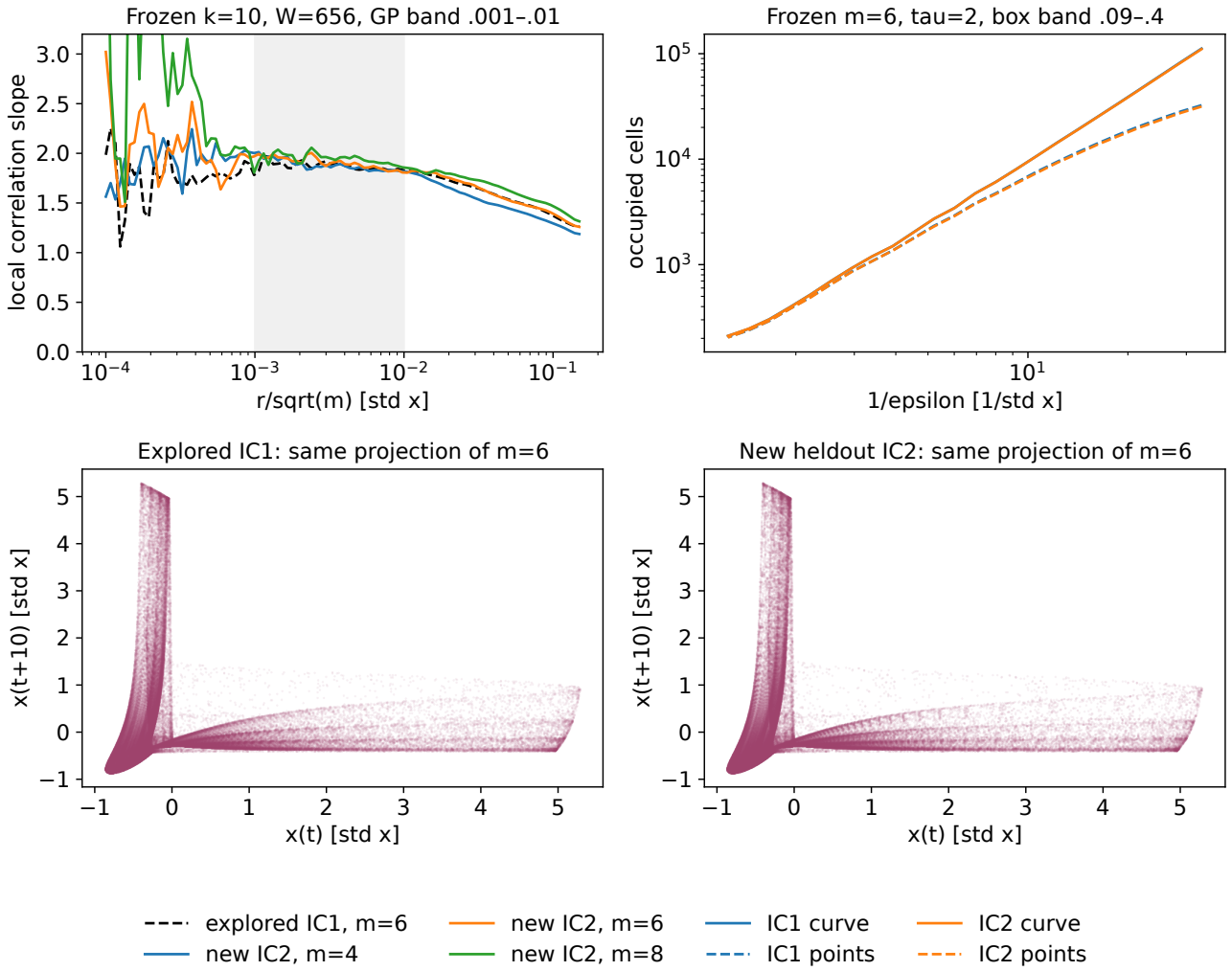


Figure 16: B4: new IC2 trajectory under frozen GP, box and projection settings

The selected m=6 result is consistent with the disclosed envelope. GP at m=8 and some point-only box fits fall outside it. Thus the primary conclusion is supported, but transferability of a narrow numerical interval is not proved. Protocol: heldout_frozen_settings.json. Both regular regimes agree across state returns, neutral model exponents and curve geometry. Chaos is supported independently of reconstruction; scalar geometry is consistent with it but does not establish an exact fractional dimension. Every finite point file also has dimension zero at arbitrarily small scales; the goal is resolved scaling of the underlying limit set, rather than that trivial file limit. The steady limit set is one point, with D0=D2=0. Normalization by zero variance, ACF and ordinary FNN are inapplicable to an established constant recording.

C. Entropy-complexity analysis

C1. Control signals

Three controls have the same N=150000 and dt=0.2: Gaussian white noise, a random permutation of the chaotic HR record, and

$$s_{qp}(t) = \sin(2\pi t/T_0) + 0.5\sin(2\pi\sqrt{2}\,t/T_0), \qquad T_0 = 20.127872376.$$

T0 is the full regular-spiking period. Shuffling preserves the exact amplitude multiset and destroys temporal order. The quasiperiodic example has incommensurate frequencies and is not classified as noise merely because it never repeats exactly.

C2. Ordinal probabilities, normalization and theoretical bounds

Bandt-Pompe patterns order $(u_i, u_{i+q}, \dots, u_{i+(\ell-1)q})$ by amplitude. Window starts advance by one sample, and all K=ell! categories, including absent ones, enter the distribution. Lehmer ranks encode permutations. Exact ties use stable temporal-index order; no random jitter is added. Tie fractions are retained. A constant record gives one convention-dependent pattern and is not interpreted as an ordinal characterization of the equilibrium.

$$S(P) = -\sum_{p_j>0} p_j \ln p_j, \quad H = S(P)/\ln K,$$

$$J = S((P+P_e)/2) - S(P)/2 - \ln K/2,$$

$$J_{max}(K) = \ln 2 + \tfrac{1}{2}\ln K - \frac{K+1}{2K}\ln(K+1), \quad C = HJ/J_{max}.$$

J is maximal at a point mass, giving this normalization. Uniform and point-mass limits are (H,C)=(1,0) and (0,0). The unit square is not the attainable region. Every H-C comparison shows theoretical minimum and maximum curves for the full alphabet K=ell!, with K=24 or 120:

$$P_{min}(p) = \left(p, \frac{1-p}{K-1}, \dots, \frac{1-p}{K-1}\right), \quad 1/K \le p \le 1,$$

$$P_{max,n}(p) = \left(p, \underbrace{\frac{1-p}{n-1}, \dots, \frac{1-p}{n-1}}_{n-1}, \underbrace{0, \dots, 0}_{K-n}\right), \quad n = 2, \dots, K, \quad 0 \le p \le 1/n.$$

Both families use the same full-K normalization as the observations, following Martin, Plastino and Rosso (2006). These are distributional bounds, not a noise/chaos decision boundary. The saved curves are `data/complexity_bounds_K24.npz` and `data/complexity_bounds_K120.npz`. Tests cover endpoints, support-size vertices, binary collapse, random distributions, all 24 length-four permutations, ties, monotone sequences, increasing-transform invariance and long IID controls. Independent tests also cover Jacobians, RK4, events, GP pair exclusion, analytic-circle dimensions and cell traversal.

C3. Regimes, consecutive segments and current variation

Every primary signal uses ell=5,q=10,dt=0.2: element lag 2 and ordinal span 8. No signal-specific tuning is used in this comparison.

signal	H	C
spiking	0.343953	0.277995
bursting	0.353274	0.279879
chaotic	0.394139	0.301186
white_noise	0.999937	0.000113417
shuffled	0.999938	0.000111083
quasiperiodic	0.556953	0.381531

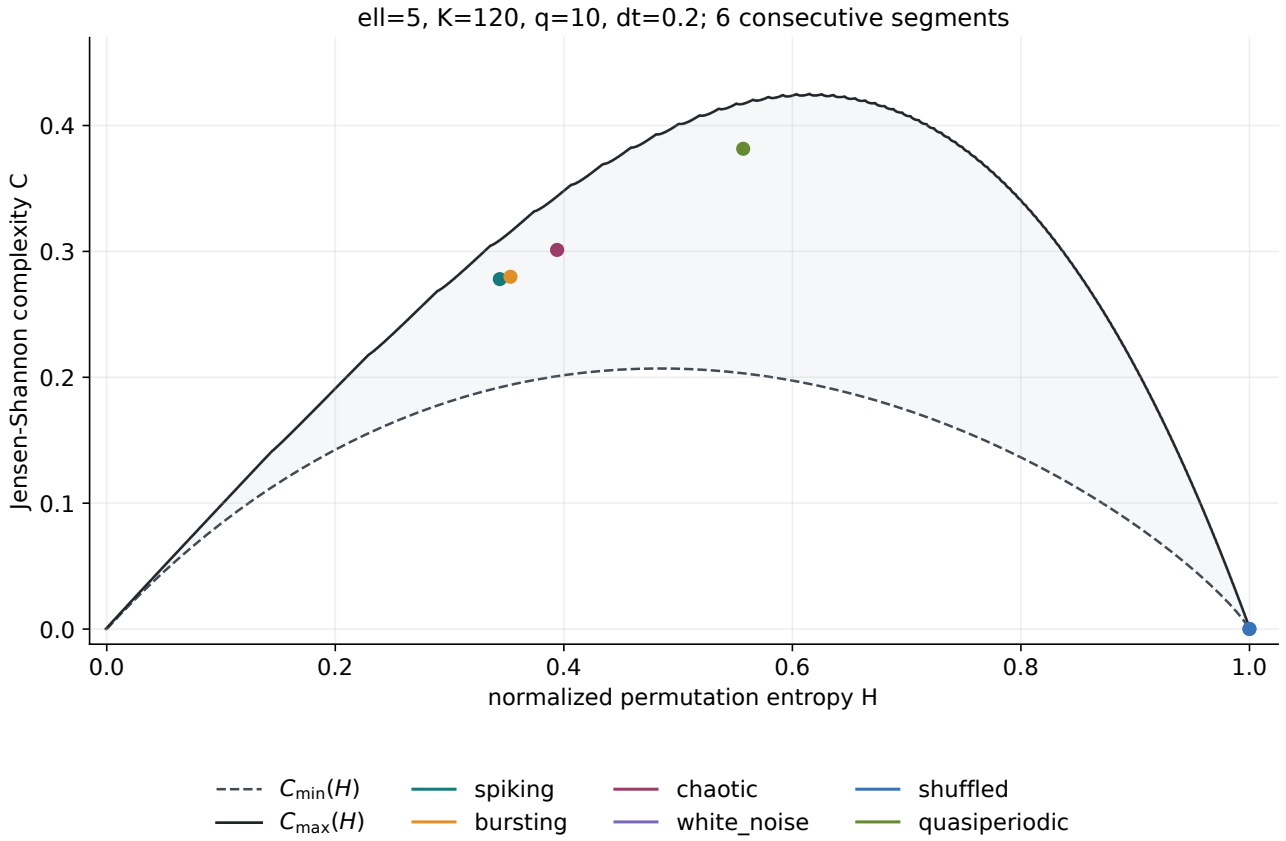


Figure 17: C3: common settings and K=120 bounds; faint points are six consecutive segments

signal	H mean	H sd	H min	H max	C mean	C sd
bursting	0.353177	0.000784405	0.352274	0.354329	0.279823	0.000490236
chaotic	0.394071	0.00110219	0.392742	0.395469	0.301181	0.000434927
quasip eriodic	0.557017	0.000184693	0.556735	0.557249	0.381561	9.170e-05
shuffled	0.999561	0.000102107	0.999375	0.999659	0.000785147	0.000182788
spiking	0.343972	0.000142813	0.34372	0.344158	0.278008	7.796e-05
white_noise	0.999498	3.406e-05	0.999462	0.999545	0.000899175	6.265e-05

These standard deviations describe consecutive segments, rather than assuming independence of overlapping ordinal windows. Noise and shuffle approach (1,0) despite different amplitude distributions. Neuronal recordings retain structure. Periodic bursting can have higher complexity than chaos at some delays; neither H nor C is a universal indicator of a positive Lyapunov exponent. For all 31 currents in the refined A3 region, H(I),C(I) and the ordered H-C path are computed:

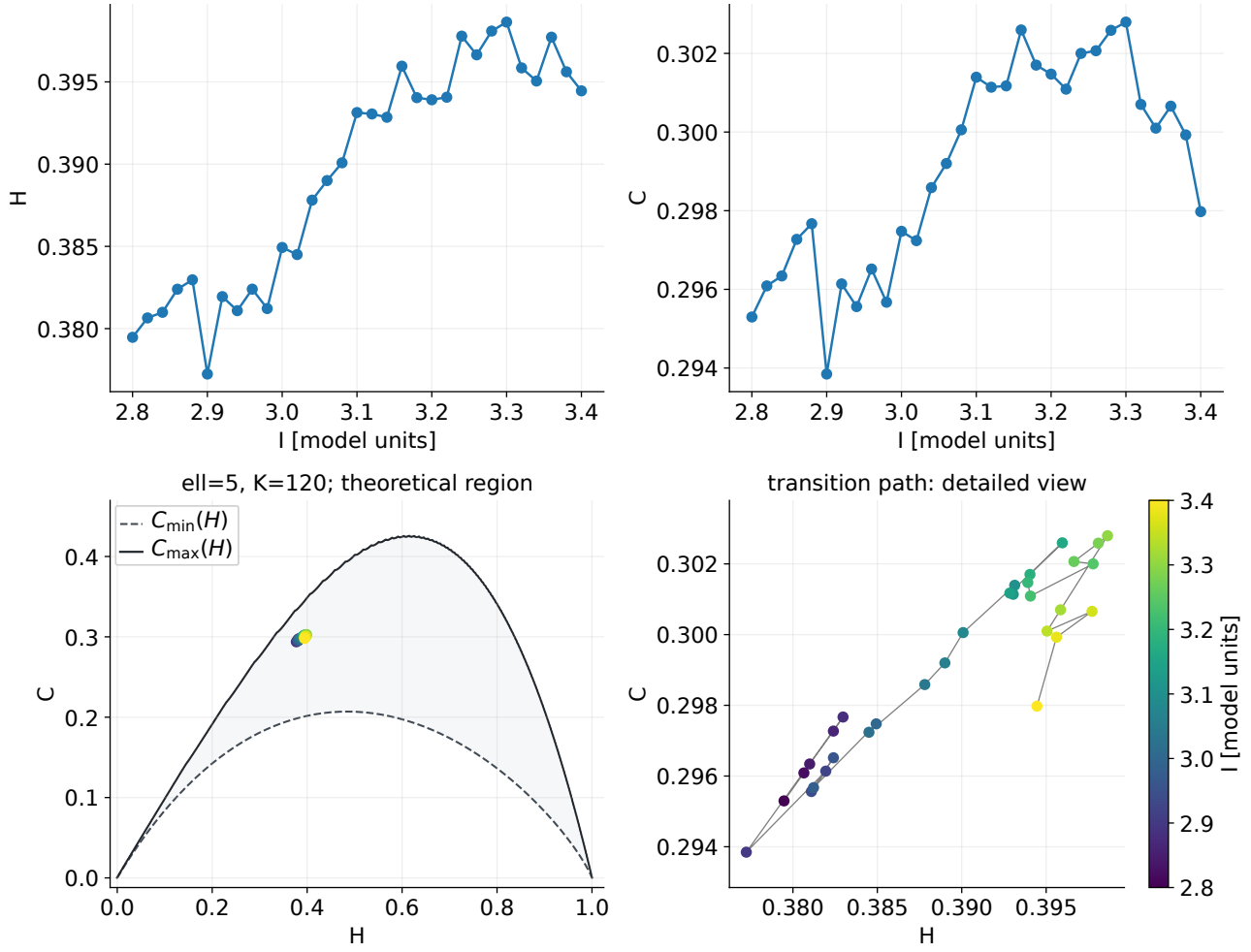


Figure 18: C3: H(I), C(I) and a 31-current H-C path at ell=5,q=10

Periodic windows and nearby irregular regimes may project close together. H-C path smoothness does not exclude a dynamical bifurcation; interpretation uses the maxima and ISI diagrams as well. Data: entropy_transition.csv.

C4. Analysis parameters and matched-scale sampling ablation

The broad study has 540 combinations: ell=4,5; q=1,5,10,25,50; observation steps 0.2,0.4,1; and 10%,50%,100% record lengths. Decimation reuses the saved trajectories and noise arrays. The physical lag q*dt is retained explicitly.

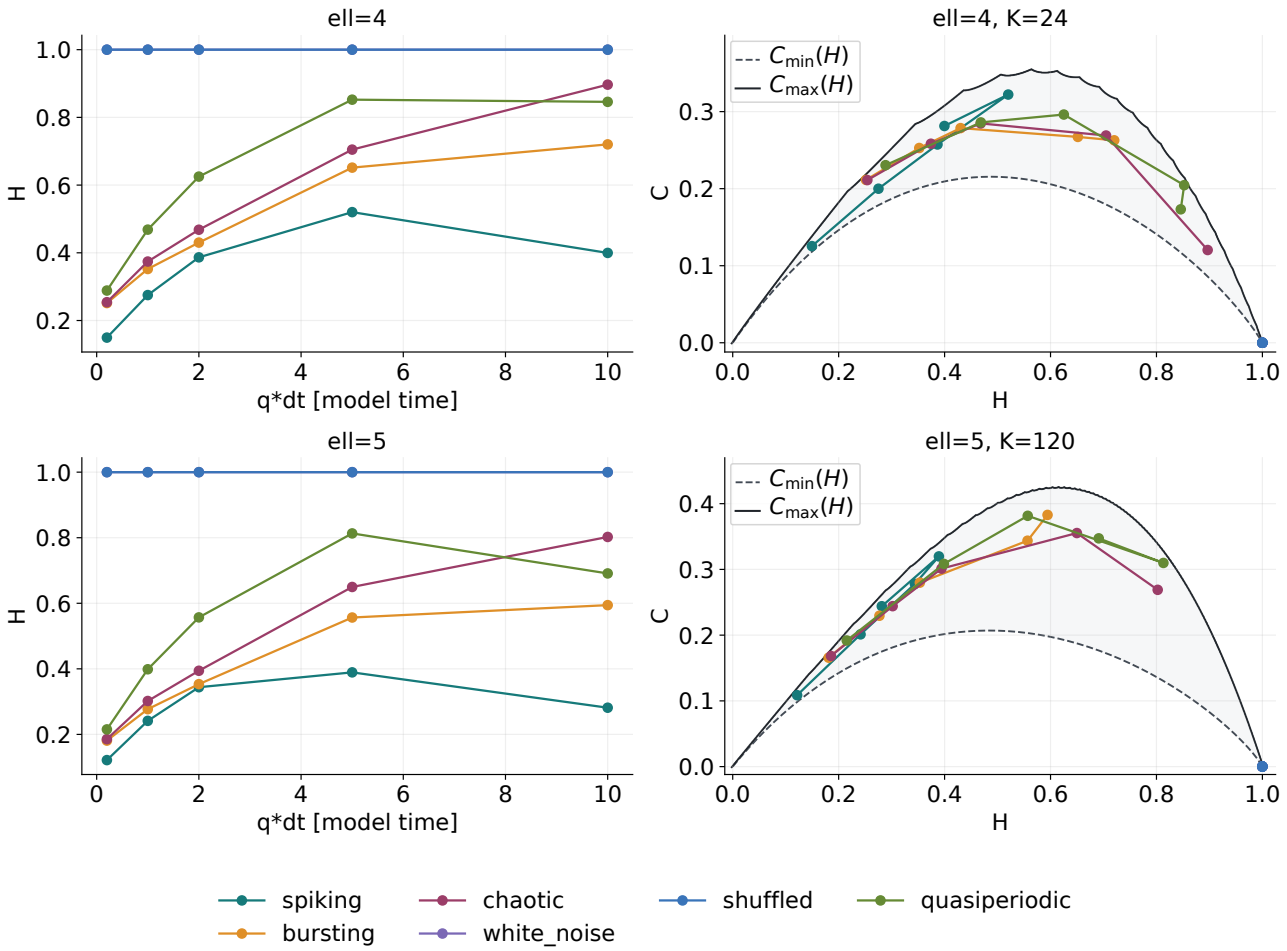


Figure 19: C4: ordinal delay and pattern-length sensitivity with the correct bounds

signal	patterns	occupied	patterns per possible	ties
spiking	149960	20	1249.67	0
bursting	149960	26	1249.67	0
chaotic	149960	29	1249.67	0
white_noise	149960	120	1249.67	0
shuffled	149960	120	1249.67	0
quasiperiodic	149960	30	1249.67	0

A primary 25000-sample segment has approximately 24960 windows, well above 120. The shortest decimated cases have much smaller windows-per-category ratios, recorded explicitly. Absent patterns in deterministic signals are not automatically sampling errors. Overlap and dependence mean that N/K alone cannot prove accuracy. All three neuronal pairs are compared using six segments: each cloud radius is its maximum segment distance from its segment mean. A positive separation margin means the descriptive circles do not intersect on these records, not a confidence probability or a guarantee for a new source.

ell	q	dt	first	second	distance between segment means	sum max segment radii	segment discs disjoint
4	1	0.2	spiking	bursting	0.133409	0.00121976	True
4	1	0.2	spiking	chaotic	0.135204	0.000935148	True
4	1	0.2	bursting	chaotic	0.00257123	0.00111368	True
4	10	0.2	spiking	bursting	0.0484421	0.00167793	True
4	10	0.2	spiking	chaotic	0.086091	0.00202976	True
4	10	0.2	bursting	chaotic	0.0385829	0.00293878	True

ell	q	dt	first	second	distance between segment means	sum max segment radii	segment discs disjoint
5	10	0.2	spiking	bursting	0.00938192	0.00164363	True
5	10	0.2	spiking	chaotic	0.0551986	0.00170861	True
5	10	0.2	bursting	chaotic	0.0461357	0.00278026	True
5	50	0.2	spiking	bursting	0.342312	0.000553498	True
5	50	0.2	spiking	chaotic	0.518866	0.0271353	True
5	50	0.2	bursting	chaotic	0.232862	0.0273668	True

The largest short-to-full chaotic-record changes are:

signal	ell	q	dt	HC drift short to full	short N	short patterns per possible
chaotic	5	25	1	0.0444563	3000	24.1667
chaotic	5	10	1	0.0422044	3000	24.6667
chaotic	5	50	1	0.0404188	3000	23.3333

The matched-scale ablation holds the physical interval [0,30000) fixed while using dt=0.2,0.4,1,2 and q=10,5,2,1, preserving q*dt=2. Its control holds q=10 and changes physical lag up to 20. Both arms save counts and six segments. Across the three neuronal signals at ell=5, maximum absolute H drift is 0.000720 in the matched arm versus 0.462466 in the fixed-q arm. Much of the broad sampling sensitivity therefore reflects a changed time scale. Decimation still reduces sample count and can create aliasing at coarser steps; this is not universal invariance.

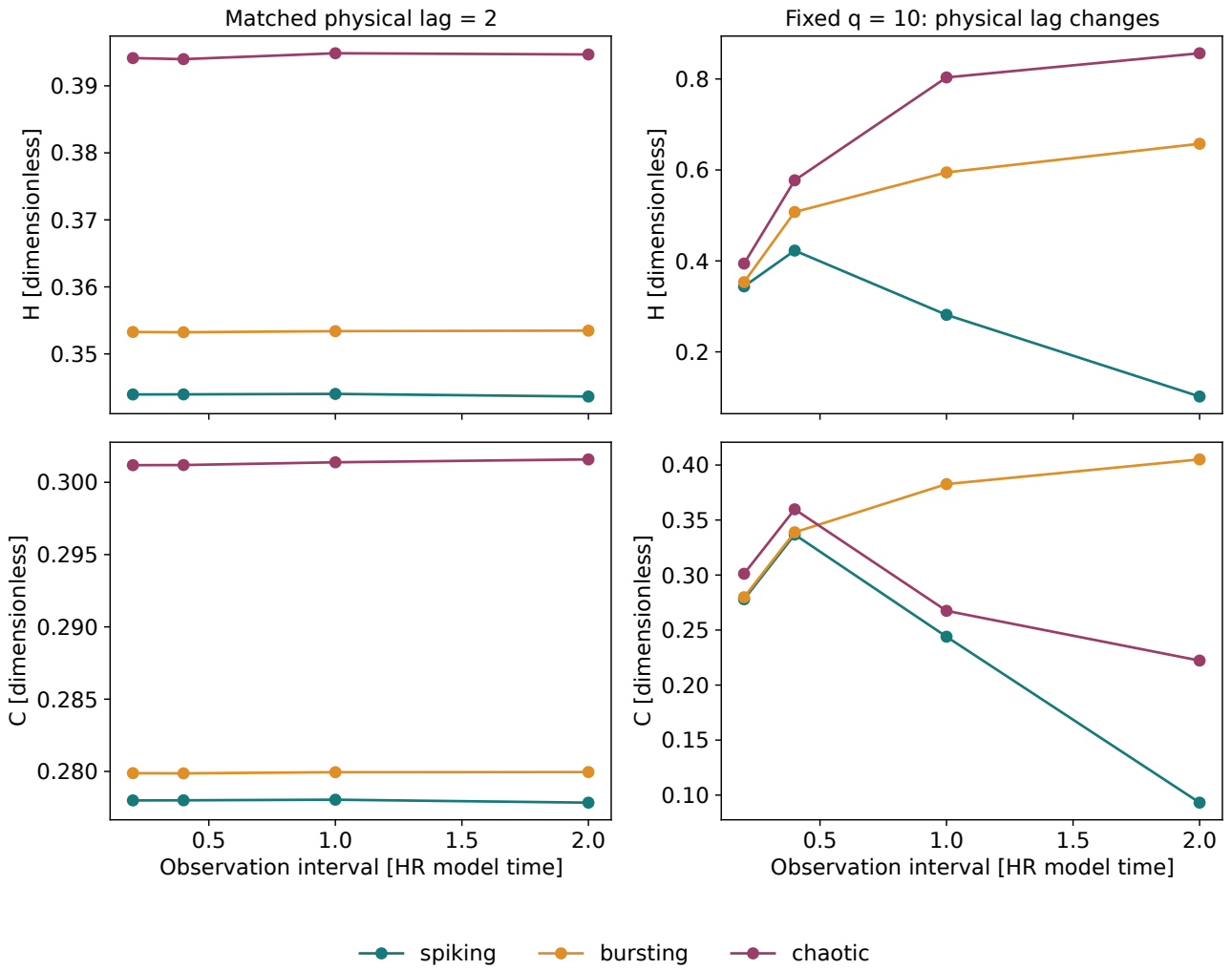


Figure 20: C4: matched physical delay versus the changing-delay control

Data: entropy_sensitivity.csv, ablation_sampling_comparison.csv.

C5. Measurement noise, paired Monte Carlo and noisy dimensions

After solving the ODEs, independent Gaussian measurement noise is added to regular spiking and chaotic records: $u_n^{obs} = u_n + \eta\sigma_u\xi_n$, eta=0,0.01,0.05,0.1,0.2. Each source's clean population-standard-deviation convention (ddof=0) sets sigma. The original experiment uses five realizations at every nonzero level and settings (ell,q)=(4,10),(5,10),(5,50). The clean level is not duplicated as independent deterministic evidence.

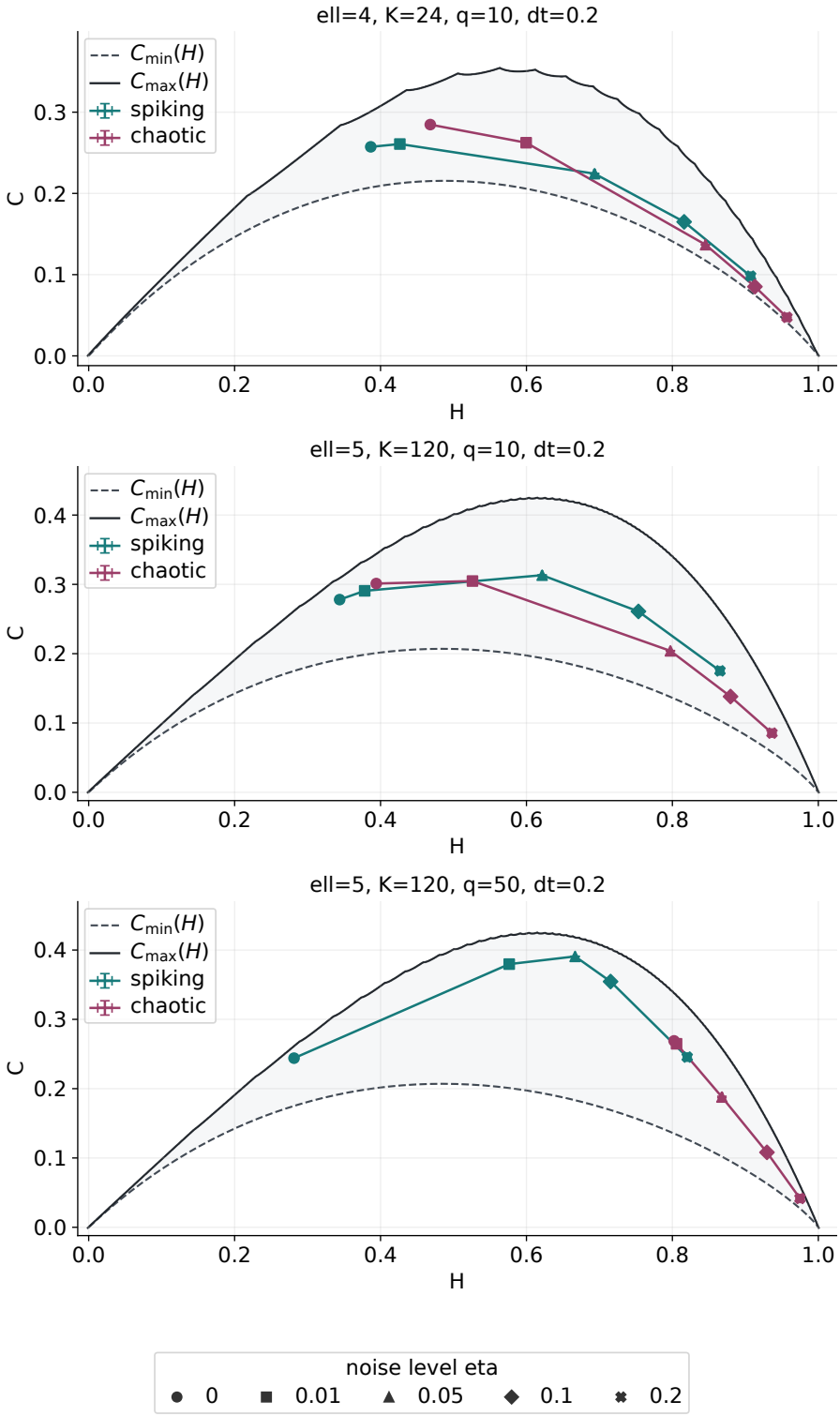


Figure 21: C5: original five-realization noise study; markers encode eta and bars show SD

Detailed numerical evidence: Appendix A.1. All columns and original row identifiers are retained.

Small noise can reorder nearby amplitudes at short delays. At the same eta, the two selected sources remain farther apart than the five-realization spread, but distance changes nonmonotonically. Unknown noise magnitude still makes interpretation of one H-C point ambiguous. Dominant IID noise drives both sources toward (1,0). At q=50, physical lag 10 is near half the spiking period; it is particularly sensitive to perturbations of almost repeated values, so larger q does not ensure robustness.

A separate supplement uses 30 independent Gaussian vectors of length 150000. Within a replicate, the same vector is used for both sources, every eta and ordinal setting (common random numbers). Replicates are independent. All vectors, their pre-draw PCG64 states and ordinal counts are retained; observed arrays can be reconstructed exactly. The original five realizations are preserved. For one statistic per replicate, mean, sample SD and $MCSE = s_g/\sqrt{30}$ are reported. A joint 10000-resample percentile bootstrap resamples complete replicate tuples, using the same indices for contrasts. Intervals are pointwise approximate 95% intervals conditional on these fixed clean trajectories. They exclude model, between-neuron and new-clean-trajectory uncertainty and are not simultaneous bands. At eta=0, Monte Carlo uncertainty is exactly zero.

The centroid distance $d_\eta = \|E(H,C)_{chaos} - E(H,C)_{regular}\|_2$ is recomputed in each bootstrap sample. It is distinguished from mean within-pair distance, which depends on the common-noise coupling. At ell=5,q=10:

eta	centroid distance	centroid mcse delta method	centroid ci low	centroid ci high
0	0.055285	0	0.055285	0.055285
0.01	0.148206	0.000162437	0.147884	0.14852
0.05	0.207849	0.00021218	0.207444	0.20825
0.1	0.175897	0.000207584	0.175495	0.176307
0.2	0.115362	0.000213471	0.114962	0.115771

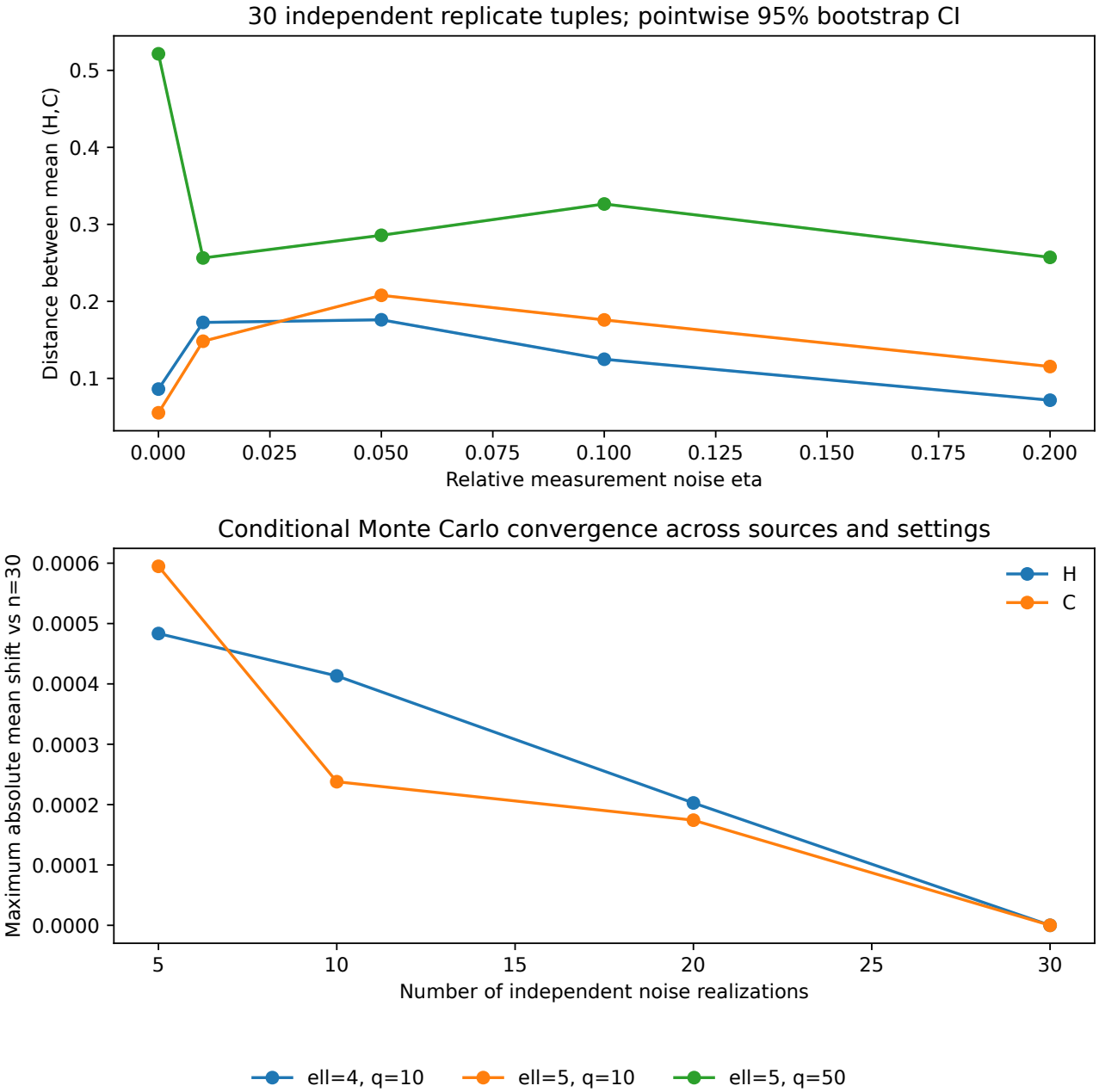


Figure 22: C5: centroid-distance intervals and Monte Carlo mean convergence

Intervals are narrower than plot markers; their numerical values are tabulated. The nonmonotonic change is much larger than conditional Monte Carlo error. q=50 follows a different path. Paired MCSE includes source covariance; its ratio to the naive unpaired estimate ranges from 0.791-1.021, so pairing need not reduce variance. Maximum H,C mean shifts relative to n=30 are:

n	maximum absolute mean shift vs n30
5	0.000594819
10	0.000413237
20	0.000202762
30	2.220e-16

These errors describe noise-ensemble means, not accuracy for classifying an individual record. Full means, SD, MCSE and intervals: noise_statistics_summary.csv, noise_statistics_paired_contrasts.csv. For eta=0.05, realization 0 of the original regular-signal

experiment, GP is repeated at the same lag and W, m=2-6:

eta	m	scale	estimate	local min	local max
0.05	2	clean_range	1.80701	1.4472	1.96552
0.05	2	alternative_range	1.62741	1.35806	1.81804
0.05	3	clean_range	2.56319	1.82083	2.92506
0.05	3	alternative_range	2.18264	1.66013	2.58269
0.05	4	clean_range	3.30867	2.12756	3.86419
0.05	4	alternative_range	2.7038	1.87393	3.34316
0.05	5	clean_range	3.98171	2.31998	4.79291
0.05	5	alternative_range	3.11432	1.99287	4.0307
0.05	6	clean_range	4.69423	2.52009	5.83377
0.05	6	alternative_range	3.56047	2.11764	4.74438

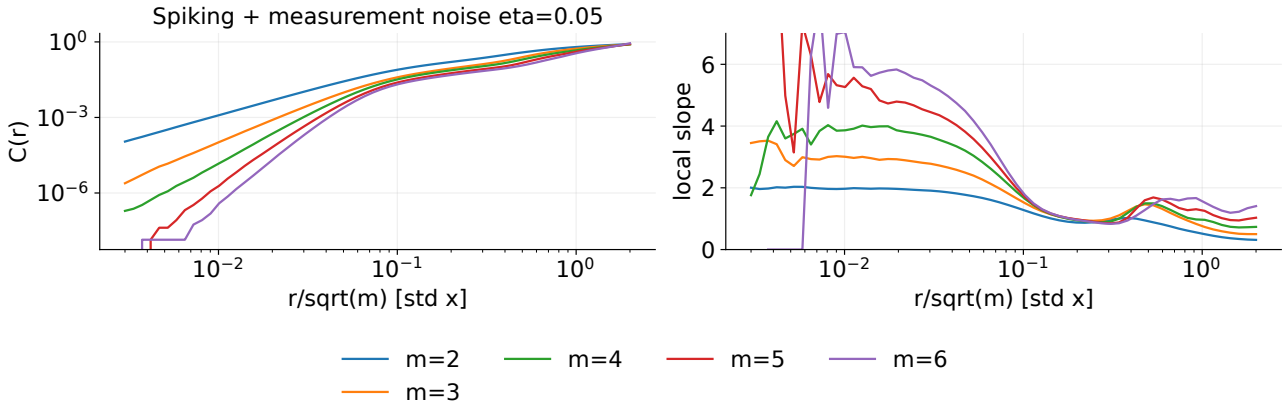


Figure 23: C5: noisy-cycle correlation sums and local slopes

Increasing small-scale slopes with m and loss of a common plateau do not mean the source ODE became chaotic. Independent continuous measurement noise produces a full-dimensional delay-space distribution at sufficiently small scales; intermediate scales retain the curve's influence. A high dimension estimate alone cannot separate noise from deterministic chaos.

D. Overall synthesis and critical discussion

The summary uses ell=5,q=10,dt=0.2 and h=0.01 model exponents. Periodic finite-time exponents are interpreted as consistent with zero. Dimension intervals are sensitivity ranges, not confidence intervals. The chaotic row uses revised m=6,k=10; point and polygonal capacity fits are distinguished. Regular D0 uses m=4 and four grid shifts. The noisy row's H,C are means of the original five realizations. Not estimated is distinct from zero, and not applicable is distinct from an unsuccessful estimate.

Detailed numerical evidence: Appendix A.2. All columns and original row identifiers are retained.

For these synthetic sources, the combined evidence distinguishes the selected regular activity, deterministic chaos and a regularly generated signal with measurement noise. A universal H-C threshold is not established.

1. Model equations provide direct evidence: an attracting equilibrium, full-state periodic return, or a consistently positive exponent on a bounded trajectory. Such evidence is unavailable for an arbitrary recording without a known model.
2. Reconstruction identifies curve geometry for regular signals and more complex chaotic geometry. Delay, coverage, temporal exclusion and resolved scales affect

its estimates. Fractional limiting values are less secure than the curve/noncurve distinction. Independent held-out results expose residual m dependence.
3. Ordinal statistics efficiently reveal ordering destroyed by shuffling and separate the selected clean neuronal and IID examples. They depend on observation scale and are not in one-to-one correspondence with exponents or dimensions.
4. Measurement noise changes ordinal order and small-scale dimension, with non-monotonic class-centroid distance. Multiple delays, controls and independent dynamical verification are necessary; narrow Monte Carlo intervals do not remove systematic bias.

The methods measure different objects: flow instability, measure/support geometry at finite spatial scales, and ordering probabilities at a chosen temporal scale. Discrepancies are therefore analyzed rather than hidden. Limits include finite observation time, unresolved narrow parameter windows, restricted reconstruction clouds, uncertain scaling limits, observation aliasing and a finite set of synthetic sources. The study neither proves every bifurcation nor guarantees error-free chaos detection in noisy experiments. Further classifier claims would require a separate design and genuinely new source populations. In particular, this study does not estimate a false-chaos probability on independent new recordings. The 30 measurement-noise replicates condition on fixed clean trajectories; the IC2 hold-out tests dimension choices, not classification sensitivity or specificity. Thus, “distinguishable” refers to the selected examples and stated parameter settings.

References

Reconstruction, FNN, temporal exclusion and dimension diagnostics follow the primary methodological sources below. The assignment has numbered citations without a bibliography; the actual primary sources used here are listed explicitly rather than assigning invented numbers.

- Hindmarsh and Rose (1984), A model of neuronal bursting using three coupled first order differential equations.
- Grassberger and Procaccia (1983), Characterization of Strange Attractors.
- Theiler (1986), Spurious dimension from correlation algorithms applied to limited time-series data.
- Kennel, Brown and Abarbanel (1992), Determining embedding dimension for phase-space reconstruction using a geometrical construction.
- Fraser and Swinney (1986), Independent coordinates for strange attractors from mutual information.
- Bandt and Pompe (2002), Permutation Entropy: A Natural Complexity Measure for Time Series.
- Rosso et al. (2007), Distinguishing Noise from Chaos.

Section 5. Hodgkin-Huxley and Izhikevich comparison

The supplementary question is which activity characteristics survive replacement of a conductance-based HH reference by a fixed two-variable Izhikevich model. The models use the prescribed parameters and initial states. A single positive affine input mapping is calibrated on designated DC currents, then frozen for all subsequent comparisons. The evidence concerns individual synthetic neurons in the investigated input range; network behavior and biological validity are outside the experiment. The calibration ablation uses its own HH/Izhikevich records, not HR observations.

5.1. HH reference and numerical method

All four variables (V, m_g, h_g, n_g) are integrated and retained:

$$\dot{V} = J(t) - 120m_g^3 h_g(V-50) - 36n_g^4(V+77) - 0.3(V+54.387),$$
$$\dot{q}_g = \alpha_q(V)(1-q_g) - \beta_q(V)q_g.$$

Capacitance is 1 microfarad/cm squared; conductances are 120,36,0.3 mS/cm squared; reversal potentials are 50,-77,-54.387 mV. Time is in ms, voltage in mV and J in microampere/cm squared. No temperature factor is introduced. The exact rates are

$$\alpha_m = 0.1\phi(V+40), \quad \beta_m = 4e^{-(V+65)/18},$$
$$\alpha_h = 0.07e^{-(V+65)/20}, \quad \beta_h = (1+e^{-(V+35)/10})^{-1},$$
$$\alpha_n = 0.01\phi(V+55), \quad \beta_n = 0.125e^{-(V+65)/80}, \quad \phi(z) = \frac{z}{-\operatorname{expm1}(-z/10)}.$$

Near zero, $\phi(z) = 10 + z/2 + z^2/120 - z^4/720000$, giving the finite limits alpha_m(-40)=1 and alpha_n(-55)=0.1. Initial V=-65 and gates are the prescribed stationary fractions alpha/(alpha+beta); exact values are saved in settings.json. HH spikes are positive crossings of V=0.

Both models use classical RK4 with h=0.01 ms. Spike times are localized with 24 bisections of a partial RK4 step. Uniform observations use dt=0.2 ms. Selected convergence checks repeat the entire transient from the initial state at h=0.005, then compare the first 400 observed ms:

signal	coarse h ms	fine h ms	max spike difference ms	coarse spikes	fine spikes	voltage RMS mV
HH_DC	0.01	0.005	4.1755e-06	27	27	0.00012883
Izh_DC	0.01	0.005	2.1905e-08	21	21	5.1932e-07
HH_weak	0.01	0.005	6.9024e-06	27	27	0.00012683
Izh_weak	0.01	0.005	4.6196e-09	20	20	3.5462e-08
HH_strong	0.01	0.005	2.7567e-08	20	20	5.4189e-06
Izh_strong	0.01	0.005	0.00015884	35	35	0.001136

Maximum event-time difference is 0.000159 ms. Event and trajectory checks support the chosen numerical precision; they do not convert a finite-time threshold bracket into an asymptotically exact bifurcation value.

5.2. Fixed Izhikevich model and hybrid events

$$\dot{v} = 0.04v^2 + 5v + 140 - u + I_{Izh}(t), \qquad \dot{u} = 0.02(0.2v - u).$$

At v=30 mV, $R(v,u) = (-65, u+8)$; the initial state is (-65,-13). The parameters a=0.02, b=0.2,c=-65,d=8 are never fitted. Voltage follows the assignment's mV scale; recovery and input use model units, and time is in ms. Input scales are therefore matched explicitly instead of assumed identical. The event time is located first, the state is moved to the event surface, reset is applied, and the remaining fraction of the same internal step is integrated. Reset is not delayed to the next output sample. Observations are right-continuous at the jump. Izh spikes are reset events; the different HH event convention affects phase but not a stationary firing rate and is retained explicitly.

5.3.1. Calibration and held-out validation

$$I_{HH} = J, \qquad I_{Izh} = \kappa J + \beta, \qquad \kappa > 0.$$

The objective is unweighted squared frequency error in Hz at training currents J=0,5,8,10,15,20. The rate estimator is $1000/\overline{ISI}$ with at least three observed spikes, otherwise zero, after a 1000-ms transient and over 2000 ms. A positive parameterization kappa=exp(theta) enforces the constraint. An Izh DC lookup with step 0.25 and h=0.02 initializes the search; actual h=0.01 integrations refine a Nelder-Mead solution. Validation currents J=7,9,12,18 affect neither initialization nor fitting. The fitted values are kappa=2.146108 model-input units per microampere/cm squared and beta=2.232621 model-input units. Full precision is saved in calibration.json and remains fixed afterwards.

split	J uA cm2	I Izh model units	HH Hz	Izh Hz	error Hz
train	0	2.2326	0	0	0
train	5	12.963	0	28.67	28.67
train	8	19.401	62.47	42.358	-20.112
train	10	23.694	68.324	51.513	-16.811
train	15	34.424	78.649	74.657	-3.9923
train	20	45.155	86.47	98.297	11.827
validation	7	17.255	58.327	37.795	-20.532
validation	9	21.548	65.629	46.93	-18.699
validation	12	27.986	72.919	60.72	-12.2
validation	18	40.863	83.535	88.777	5.2428

split	RMSE Hz	MAE Hz	max abs Hz
train	16.658	13.569	28.67
validation	15.391	14.168	20.532

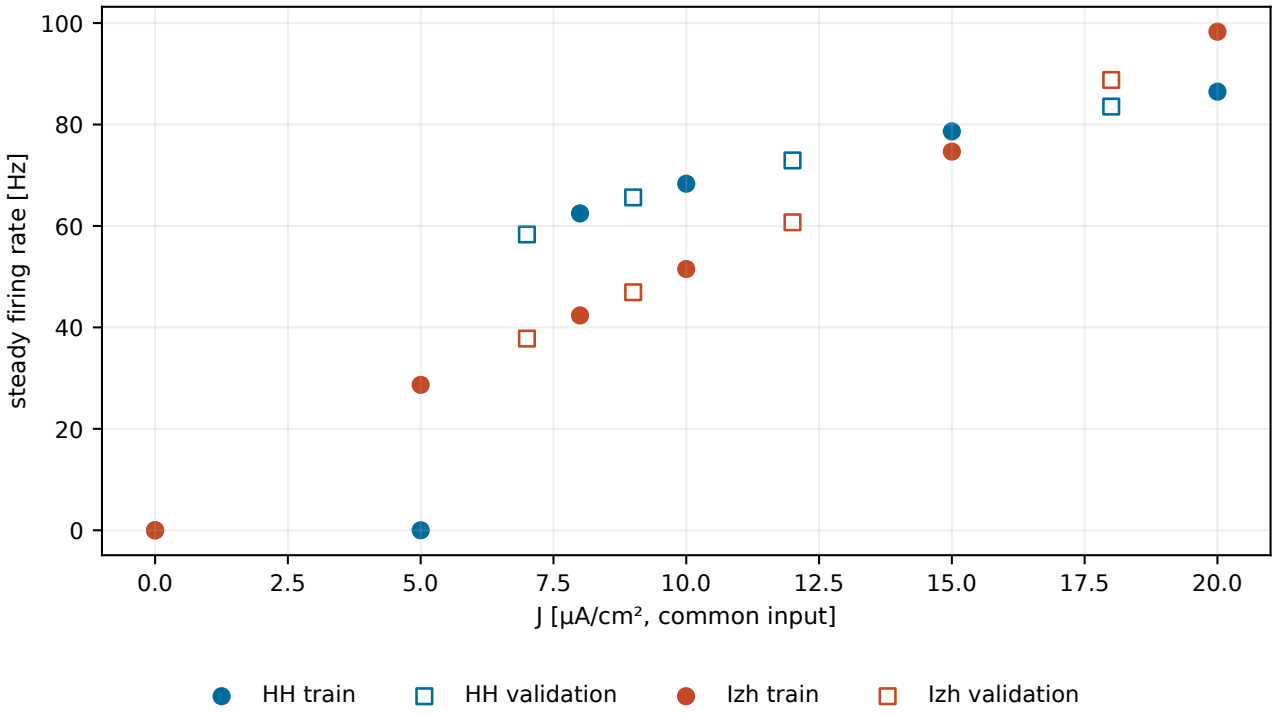


Figure 24: DC calibration; filled training and open validation markers

Ablation of the input transformation

The identity mapping kappa=1,beta=0 and the frozen fitted mapping are compared with the same Izh equations, initial state, 1000-ms transient, 2000-ms observation and h=0.02. Both branches are newly integrated; no optimization or validation retuning occurs. The rate estimator is unchanged.

split	arm	RMSE Hz	MAE Hz	max abs Hz
train	identity	36.818	31.615	46.008
train	frozen_fitted	16.658	13.569	28.67
validation	identity	44.679	44.657	46.305
validation	frozen_fitted	15.391	14.168	20.532

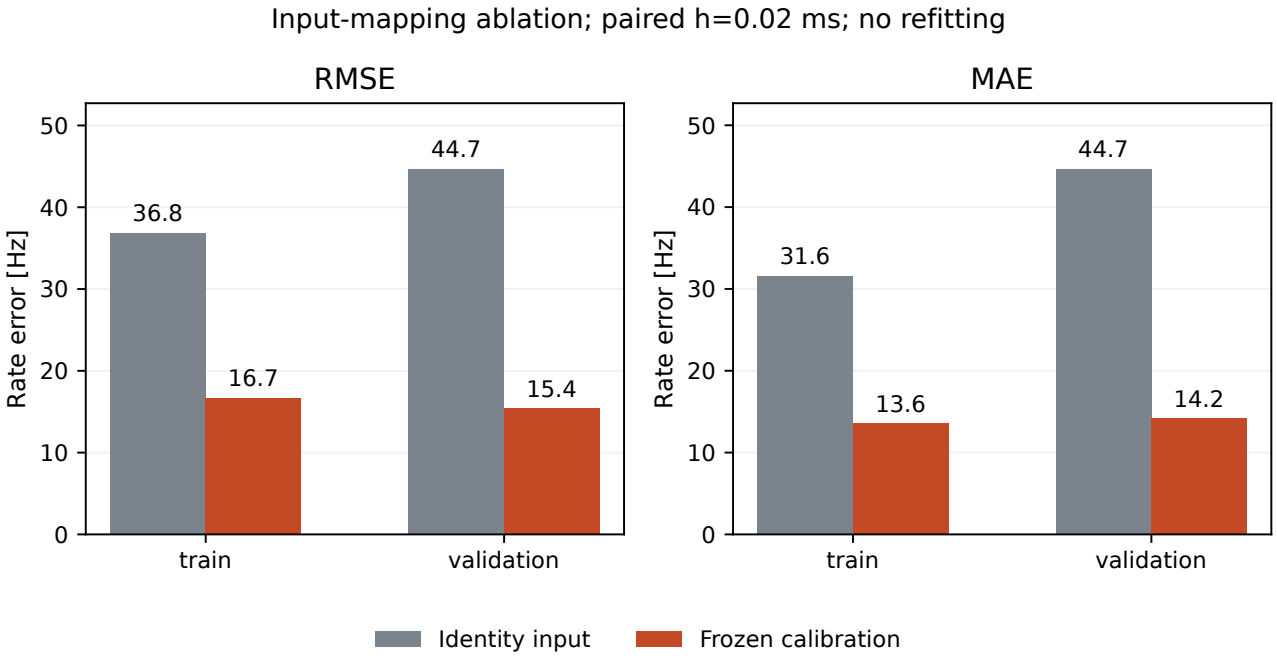


Figure 25: Effect of frozen input calibration on train and validation errors

Calibration reduces both errors but leaves substantial mismatch. The frozen h=0.02 rates differ from the original h=0.01 fitted rates by at most 4.44e-08 Hz, excluding step size as the explanation of the improvement. Paired rates and error scores retain every current. This ablates input matching, not a prescribed neuron mechanism.

Five independent differential-evolution starts on the training lookup and alternative constraints on active training points probe optimization sensitivity:

method	kappa	beta	lookup SSE Hz2	exact SSE Hz2
independent_DE_seed_1	2.1463	2.23	1664.9	—
independent_DE_seed_2	2.1463	2.23	1664.9	—
independent_DE_seed_3	2.1463	2.2299	1664.9	—
independent_DE_seed_4	2.1463	2.2299	1664.9	—
independent_DE_seed_5	2.1463	2.2298	1664.9	—
J0_active_refined	2.0356	3.81	1677.6	1707.9
J5_quiet_refined	2.9782	-11.141	2344.8	2344.8
selected_exact_train_fit	2.1461	2.2326	1664.9	1664.9

The tested starts agree near the same minimum. Requiring silence at J=5 increases training SSE. This exposes a calibration tradeoff, without proving global optimality on an unbounded parameter domain or impossibility for every other criterion. The fitted model can fire where HH remains quiet; onset mismatch is quantified separately.

Operational thresholds and transient false positives

Sustained firing is defined on the finite observation interval: every one of four equal windows must contain at least three spikes, window mean-ISI rates must vary by at most 1%, and the final silent gap must not exceed twice the median of the last eight ISIs. The rule is saved in `data/firing_criterion.json`. The basic threshold search uses transient 1500 ms, observation 3000 ms and bisection at h=0.02,0.01,0.005:

model	h ms	J lower uA cm2	J upper uA cm2	threshold mid	bracket width uA cm2	transient ms	duration ms
HH	0.02	6.2603	6.2603	6.2603	1.1444e-05	1500	3000
HH	0.01	6.2603	6.2603	6.2603	1.1444e-05	1500	3000
Izh	0.02	0.71838	0.7184	0.71839	1.9073e-05	1500	3000
Izh	0.01	0.71838	0.7184	0.71839	1.9073e-05	1500	3000

A total count of at least three spikes is insufficient: the recorded regression case fires 15 times near the beginning and then stays silent. Window and tail checks reject it:

Detailed numerical evidence: Appendix A.3. All columns and original row identifiers are retained.

Independent definitions compare first-spike onset over 200 ms without transient with sustained firing after transient 3000 and observation 6000:

model	definition	J lower uA cm2	J upper uA cm2	duration ms
HH	first_spike	2.2368	2.2368	200
HH	sustained_long	6.2603	6.2603	6000
Izh	first_spike	0.56751	0.56752	200
Izh	sustained_long	0.71838	0.7184	6000

model	short mid	doubled mid	finite time shift uA cm2
HH	6.2603	6.2603	0
Izh	0.71839	0.71839	-2.3842e-06

These are operational finite-time thresholds. Bracket width measures localization of the chosen criterion, not uncertainty in an infinite-time bifurcation. Near onset HH has a large finite firing-rate gap, whereas Izh has a much smaller one. Count rates at the upper long-observation endpoints are:

model	J upper uA cm2	upper events	duration ms	count rate Hz
HH	6.2603	301	6000	50.167
Izh	0.7184	26	6000	4.3333

Thus the comparison uses quantitative onset behavior instead of claiming that one f-I curve is necessarily mathematically continuous. The input calibration is unchanged after correcting the sustained-firing test. Independent tail checks find 17 active and 3 quiet responses among all 20 saved calibration/validation records; the active windows contain at least 14 events. The original frequency fit is not contaminated by the transient-count false positive. Data and boundary trajectories: calibration_tail_stability.csv, threshold_definition_checks.csv.

5.3.2. Periodic forcing, synchronization and transitions

$$J(t) = 10 + A\sin(2\pi t/T), \quad T \in \{10, 20, 40\}\ \text{ms}, \quad A \in [0, 15].$$

A is in microampere/cm squared. The amplitude grid is 0.5 for T=10,40 and 0.1 for the detailed T=20 study (151 values). Every point restarts from the prescribed initial state, discards 100 forcing cycles and observes 200. All state components, uniform dt=0.2 records, spike times and fixed-phase stroboscopic states X(nT) are saved. The experiment does not exclude additional coexisting attractors. Synchronization uses spikes per input cycle, ISI, full-state stroboscopic recurrence and phase resultants $R_k =$

$|\langle e^{ik2\pi t_{spike}/T}\rangle|$, k=1,2. R1 below 1 alone does not exclude multispike locking. A period p<=8 is accepted when the scaled late-state recurrence RMS is below 1e-5; state scales are (100,1,1,1) for HH and (100,30) for Izh. Label zero means that period <=8 is unresolved, not that chaos is proved.

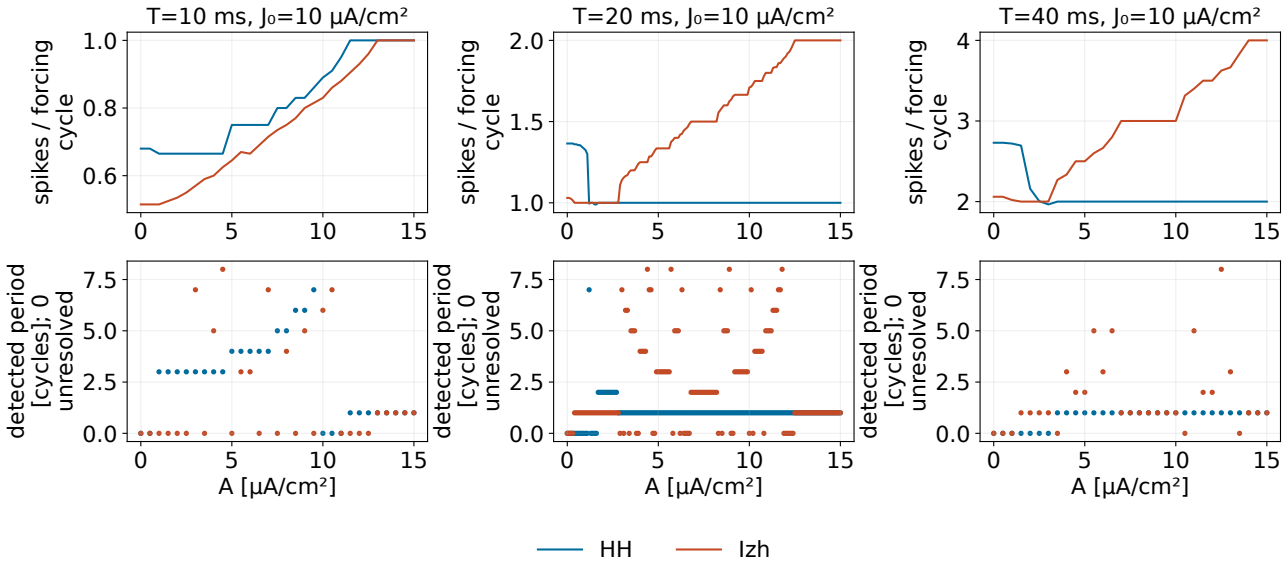


Figure 26: Responses at all three forcing periods

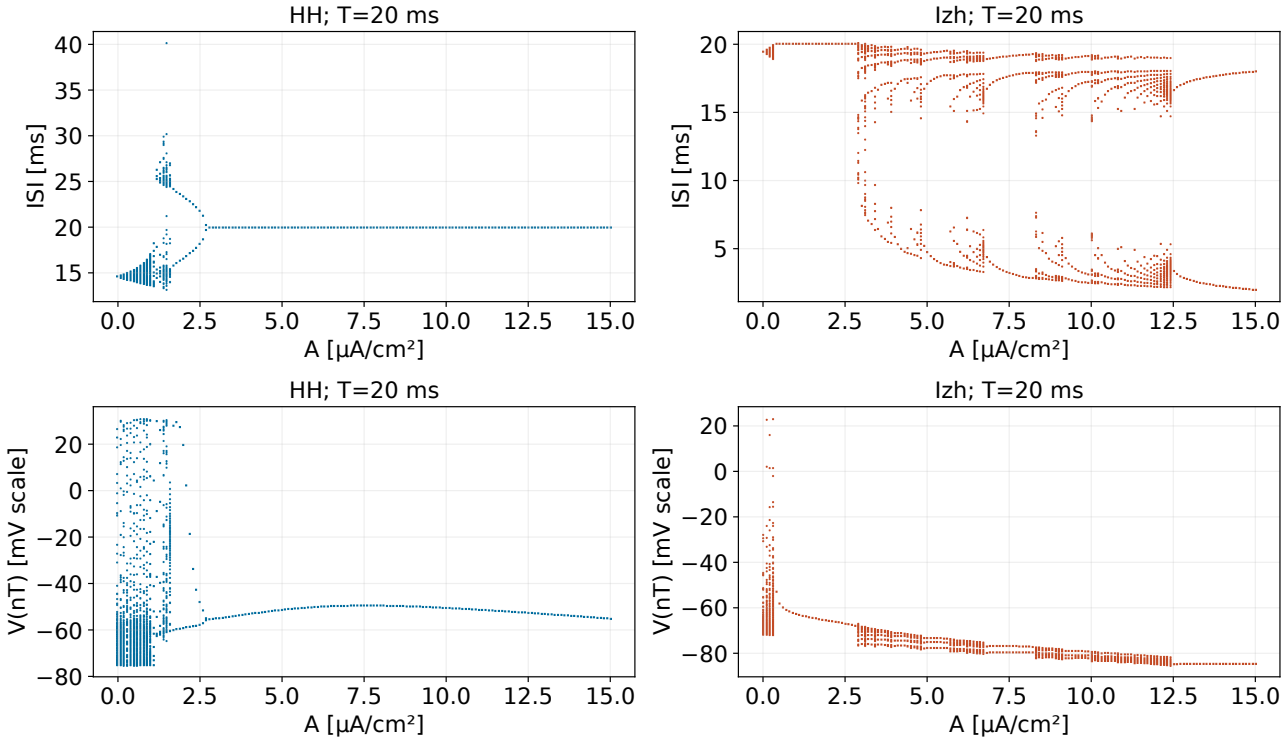


Figure 27: Detailed amplitude sweep at T=20 ms

Grid-contiguous stroboscopic-period intervals at T=20:

Detailed numerical evidence: Appendix A.4. All columns and original row identifiers are retained.

These boundaries initially have amplitude-grid resolution 0.1. An apparent period multiple alone does not identify a bifurcation type. Unresolved responses could be qua-

siperiodic, long-transient, long-period or chaotic. Convergence checks around short p7/p8 islands and locking boundaries cover 22 amplitudes, transient 16000 ms, observation 4000 ms, and h=0.01,0.005. The recurrence label matches the primary grid in 44/44 checks. Maximum spikes-per-cycle difference is 0.015; finite-time ratios are not declared exact asymptotic rotation numbers. Two boundaries are also examined with amplitude step 0.01 and h=0.005:

model	A lower	lower period	A upper	upper period
HH	2.7	2	2.71	1
Izh	0.36	0	0.37	1

These are local changes of a tolerance-based recurrence label, not rigorously identified bifurcation parameters. Full records and settings: transition_convergence.csv, transition_grid_refinement.csv. Comparison of mean spikes per forcing cycle:

T ms	A points	mean abs rate ratio difference	matching ratio abs lt 0.02
10	31	0.069032	5
20	151	0.53427	17
40	31	0.9821	1

Refined periodic solutions and hybrid stability

A stable forced response with T=20 and A>5 is selected in each model. Shooting solves $F(X) = \Phi_T(X) - X = 0$ with Jacobian $D\Phi_T - I$ using `scipy.optimize.root`. DOP853 integrates state and variational equations with rtol=2e-10,atol=1e-11,max-step=0.1 ms; tighter tolerances are also checked. On smooth segments, $\dot{M} = Df\,M$. At $g(v,u) = v - 30 = 0$, n=(1,0), $DR = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, the saltation matrix is

$$S = DR + \frac{[f^+(t, R(X^-)) - DRf^-(t, X^-)]n^T}{n^T f^-(t, X^-)}.$$

The event and reset are time independent, but both vector fields use the same actual event time and forcing phase. $M^+ = SM^-$ accounts for a perturbed reset time. DR alone is incorrect. A neutral autonomous multiplier is not required when forcing phase is prescribed.

Detailed numerical evidence: Appendix A.5. All columns and original row identifiers are retained.

Matrices and multipliers are saved in `data/orbit_HH.npz` and `data/orbit_Izh.npz`. Central finite differences perturb the full hybrid period map, including shifted reset times. Closure residuals, derivative error and spectral radius are reported; `wrong_DR_only_spectral_radius` displays the effect of omitting saltation. A radius below 1 establishes local asymptotic stability of these refined periodic solutions, rather than every point of the sweep.

Model Lyapunov exponents at fixed prescribed forcing phase

A physical-state tangent obeys $\dot{w} = Df(t, X)w$ and Izh resets use $w^+ = Sw^-$. The analytic HH Jacobian includes the removable rate limits. Joint state/tangent RK4 splits at reset events localized by 30 bisections. Every T=20 ms the tangent is normalized and its pre-normalization norm r_j is retained:

$$\hat{\lambda}_{max}(L) = \frac{1}{L}\sum_{j} \ln r_j.$$

Forcing phase is not a perturbed tangent coordinate. Hence an attracting forced cycle has a negative exponent, while an autonomous A=0 cycle retains a neutral time-shift direction. Both state and tangent discard 4000 ms. Observation times and steps vary without recalibration; states, events and cycle growth logs are saved. The following estimates are in inverse ms; the last column spans four disjoint 8-s blocks of the 32-s record.

HH weak:

h ms	4s	8s	16s	32s	8s block range
0.005	-0.0006425	-0.0002931	-0.0001446	-4.16e-05	-0.000317 ... 0.000439
0.01	-0.0006425	-0.0002931	-0.0001446	-4.16e-05	-0.000317 ... 0.000439
0.02	-0.0006425	-0.000293	-0.0001446	-4.165e-05	-0.000317 ... 0.000439

Izh strong:

h ms	4s	8s	16s	32s	8s block range
0.005	0.00418	0.002076	0.002138	0.00187	0.00148 ... 0.0022
0.01	0.00367	0.002796	0.002087	0.00191	0.00138 ... 0.0028
0.02	0.002237	0.001807	0.001552	0.0014	0.000901 ... 0.00181

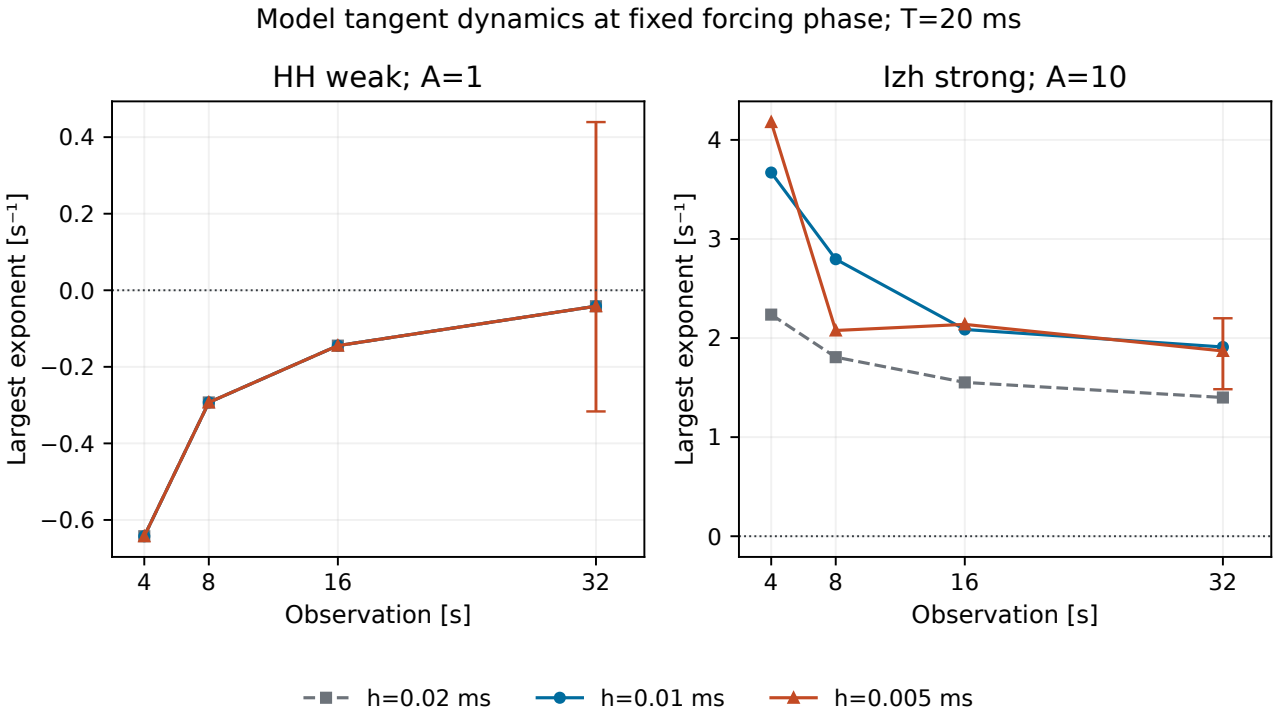


Figure 28: Model-exponent time and step convergence; bars span blocks and are not confidence intervals

The plotted units are inverse seconds. All 12 long blocks for Izh strong across three steps are positive; the minimum is 0.000901 inverse ms. This independently supports a chaotic response at A=10,T=20. Magnitudes still vary with finite duration and numerical orbit, so an exact limiting exponent is not claimed. HH weak approaches a near-zero

average with both block signs: quasiperiodicity is plausible, but a very long cycle is not excluded and an invariant torus is not proved. Longest-record selected controls at the finest available step:

signal	h ms	observation ms	lambda max per ms	quarter min per ms	quarter max per ms
HH_DC	0.01	3.2e+04	-0.0001063	-0.0004457	0.000134
Izh_DC	0.01	3.2e+04	-5.957e-05	-0.0005406	0.0003926
HH_weak	0.005	3.2e+04	-4.16e-05	-0.0003165	0.0004394
Izh_weak	0.01	8000	-0.01553	-0.01553	-0.01553
HH_strong	0.01	8000	-0.1837	-0.1837	-0.1837
Izh_strong	0.005	3.2e+04	0.00187	0.001484	0.0022

Small finite-time DC deviations from zero partly reflect endpoint phase. One-period state/tangent maps are independently checked against DOP853:

model	h ms	matrix relative error	event time error max ms
HH	0.02	1.512e-06	—
HH	0.01	9.19e-08	—
HH	0.005	5.688e-09	—
Izh	0.02	7.964e-08	2.17e-07
Izh	0.01	6.021e-09	1.385e-08
Izh	0.005	4.329e-10	8.774e-10

The refined Izh orbit agrees with log(spectral radius)/T to 1.64e-10 inverse ms. This validates the tangent solver rather than inferring chaos from ordinal entropy. See forced_lyapunov.py, forced_lyapunov.csv and forced_lyapunov_map_checks.csv.

5.3.3. Scalar observations of both models

The same inputs A=0,1,10,T=20,J0=10 are analyzed for both models: transient 2000 ms, observation 8000 ms, dt=0.2 and N=40001. Only membrane potential is passed to scalar estimators, although all variables are saved. Standardization removes amplitude scale without changing ordering; the Izh jump is not smoothed. Independent classifications are:

signal	classification
HH_DC	autonomous regular firing; neutral time-shift direction
Izh_DC	autonomous regular firing; neutral time-shift direction
HH_weak	period <=8 unresolved; near-neutral exponent; quasiperiodicity plausible, unproved
Izh_weak	stable stroboscopic period 1T; λ<0
HH_strong	stable stroboscopic period 1T; λ<0
Izh_strong	numerically chaotic: positive model exponent at three steps and in every 8-s block

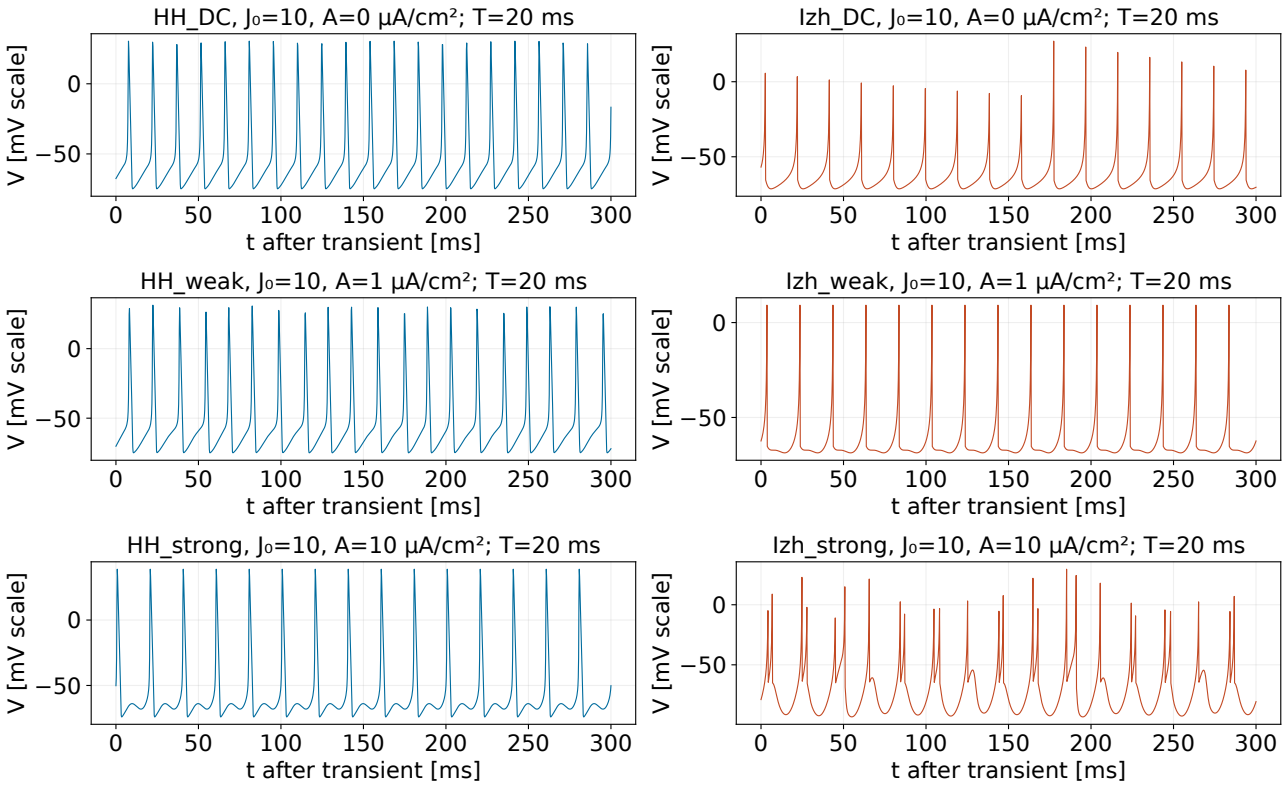


Figure 29: Selected scalar potential recordings

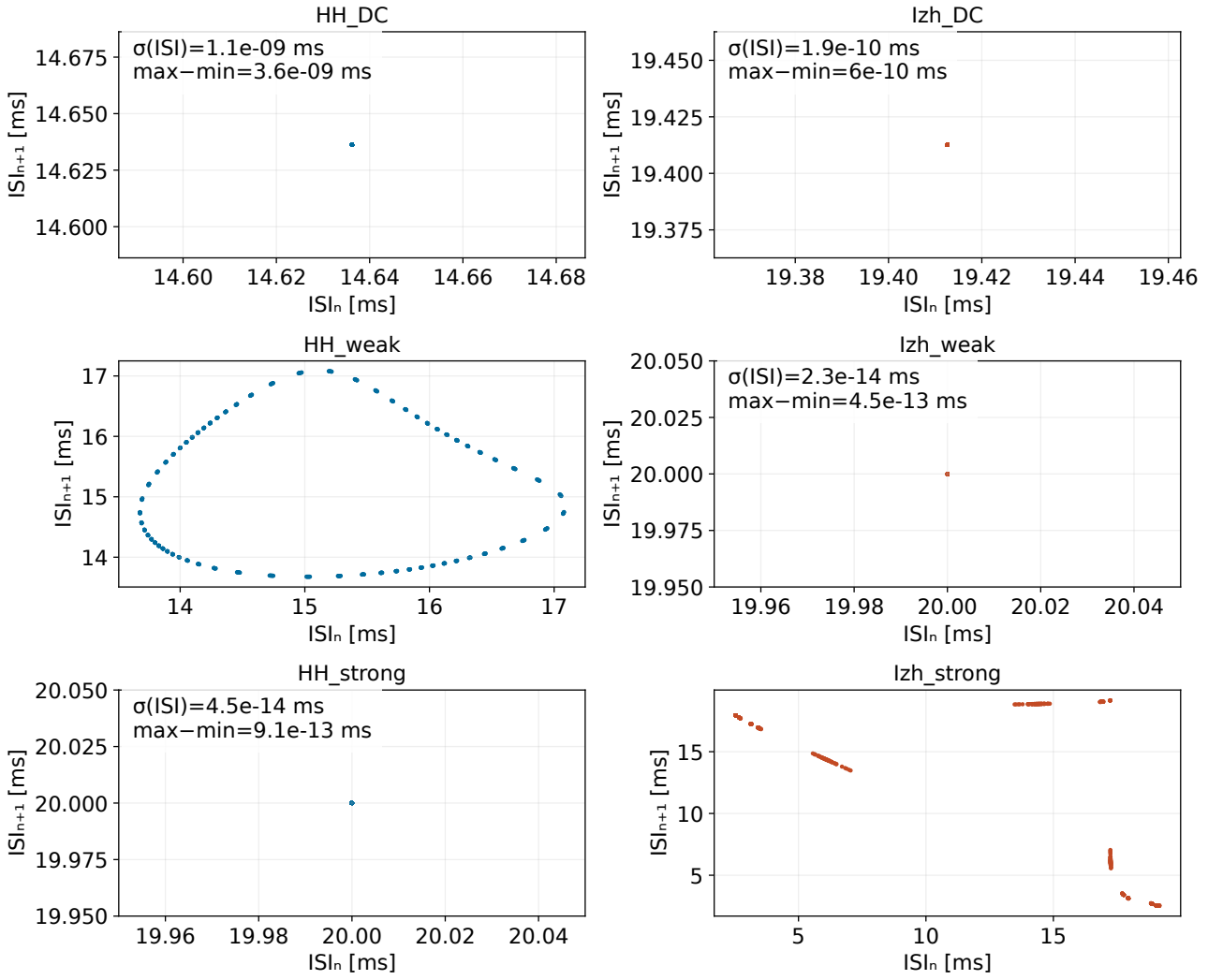


Figure 30: ISI return plots with controlled axes for almost periodic records

For ISI CV below 1e-4, return axes are fixed around the mean by +/-0.05 ms, with actual SD and range annotated, preventing numerical jitter from becoming a visible cloud through autoscaling. At A=0, stroboscopically sampling an autonomous cycle can trace a curve if its period is incommensurate with T; this does not make the continuous orbit two-dimensional.

Reconstruction and comparable delay geometry

ACF uses the first nonpositive lag; MI uses 32 equal-width bins and its first local minimum. FNN is evaluated at the individual MI lag, Theiler window 40 ms and up to 5000 deterministic reference indices, with ratio thresholds 10 and 2. Nearly repeated phases and tiny neighbor distances leave residual FNN, including 2-3% in some regular cases. FNN is a diagnostic rather than an embedding proof for a hybrid discontinuous observable.

signal	ACF first zero samples	MI first min samples	ACF zero ms	MI min ms	FNN m3	FNN m5
HH_DC	10	5	2	1	0.0308	0.0292
Izh_DC	18	10	3.6	2	0.0238	0.0232
HH_weak	10	6	2	1.2	0.0408	0.033
Izh_weak	17	3	3.4	0.6	0.27133	0.020383
HH_strong	11	11	2.2	2.2	0.070312	0.0292
Izh_strong	25	22	5	4.4	0.0602	0.0274

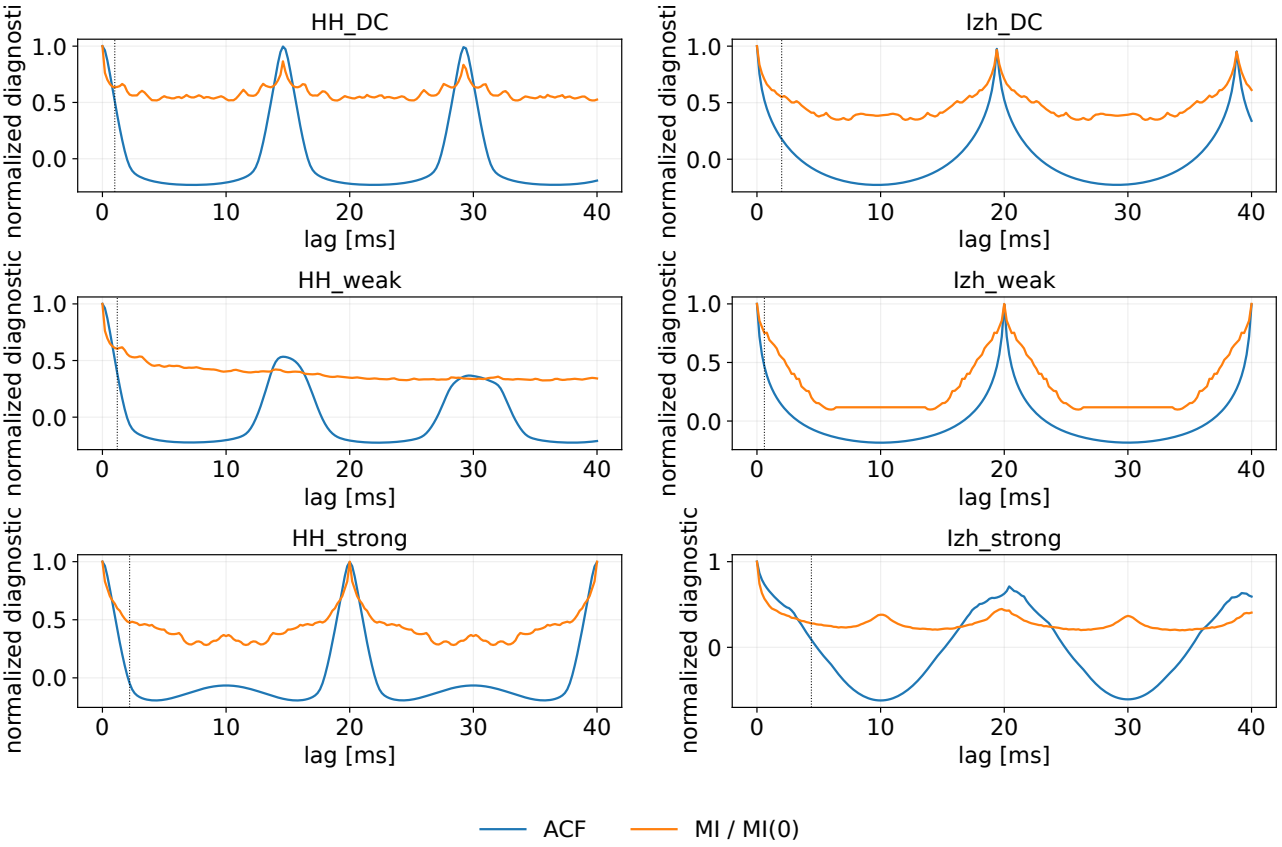


Figure 31: ACF and mutual-information diagnostics

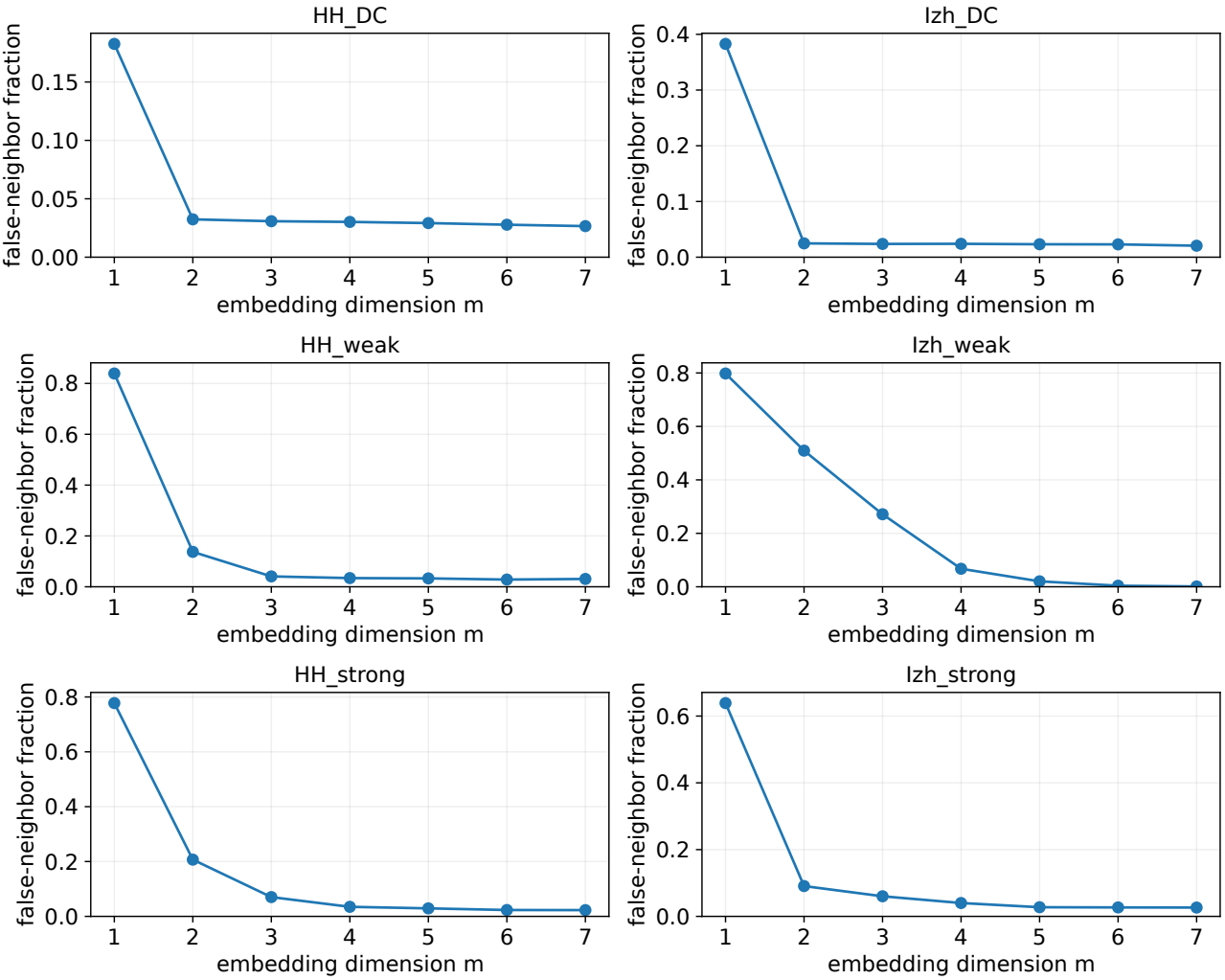


Figure 32: Individual-delay false nearest neighbors

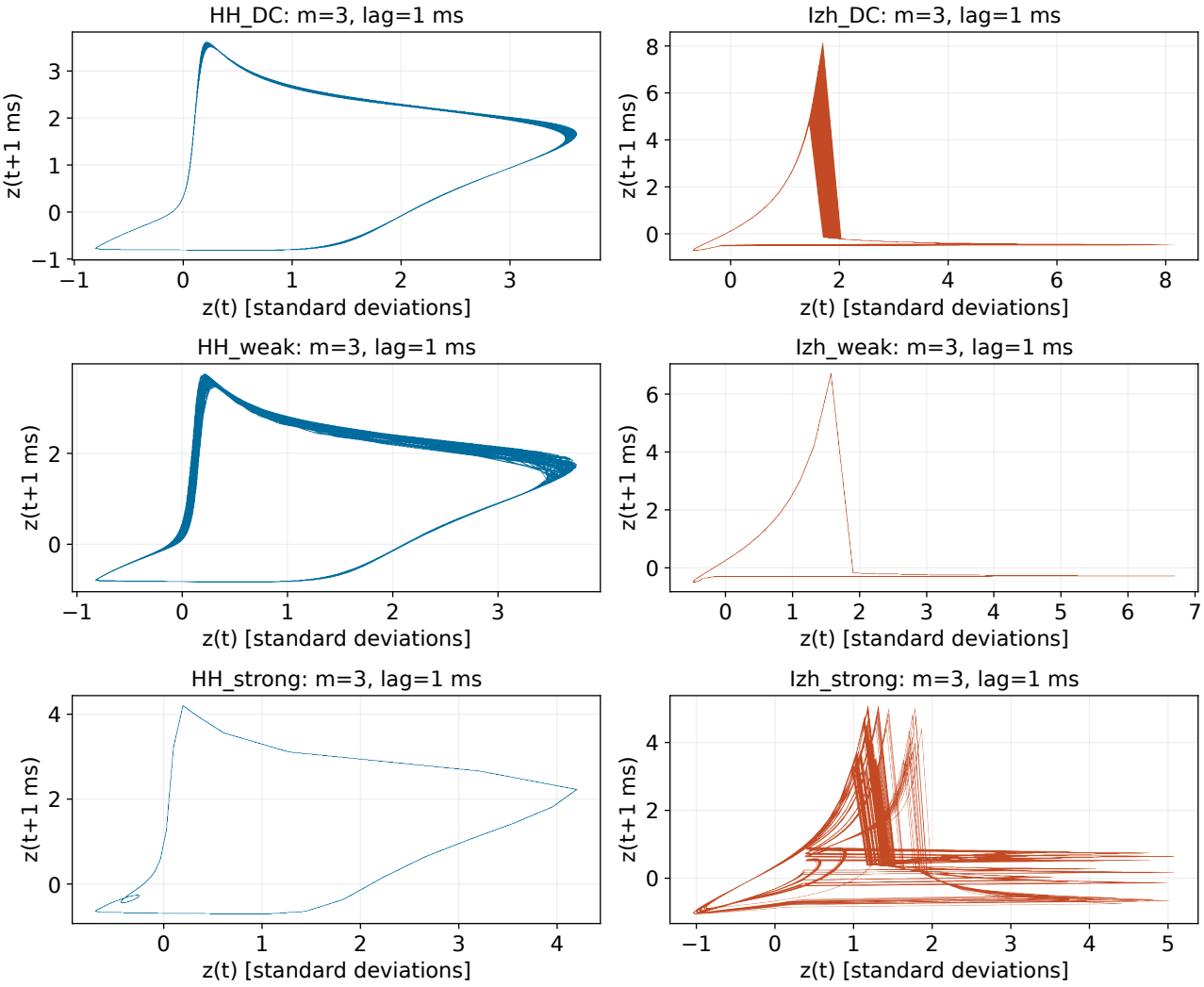


Figure 33: Scalar delay projections

The primary comparison fixes physical delay 1 ms (five samples), with individual MI and 2-ms alternatives. FNN is also evaluated at this same common delay and at the actual m=4 GP embedding:

signal	lag ms	m GP	FNN m3	FNN m4	FNN m5	FNN m7
HH_DC	1	4	0.0308	0.0302	0.0292	0.0266
Izh_DC	1	4	0.0072	0.0072	0.0068	0.0072
HH_weak	1	4	0.0358	0.0296	0.027	0.0262
Izh_weak	1	4	0.095775	0.019679	0.0060024	0.0012
HH_strong	1	4	0.060698	0.020833	0.0106	0.0092
Izh_strong	1	4	0.0076	0.0072	0.0086	0.0084

Residual FNN and scale sensitivity limit geometric interpretation. A common m=4 provides comparable geometry; it is not asserted to be a proven minimal embedding.

Correlation and capacity estimates

GP uses the same admissible-pair normalization as B, up to 3500 reference vectors, Theiler window 40 ms, and sensitivity to 0,20,80 ms windows, m=2-6, record length and delay. The fitting interval r=0.06-0.35 uses standardized Euclidean distance, at least 200 close pairs, C<0.15 and at least four scales. It is common across models, rather than chosen for a desired dimension. Local slopes are displayed.

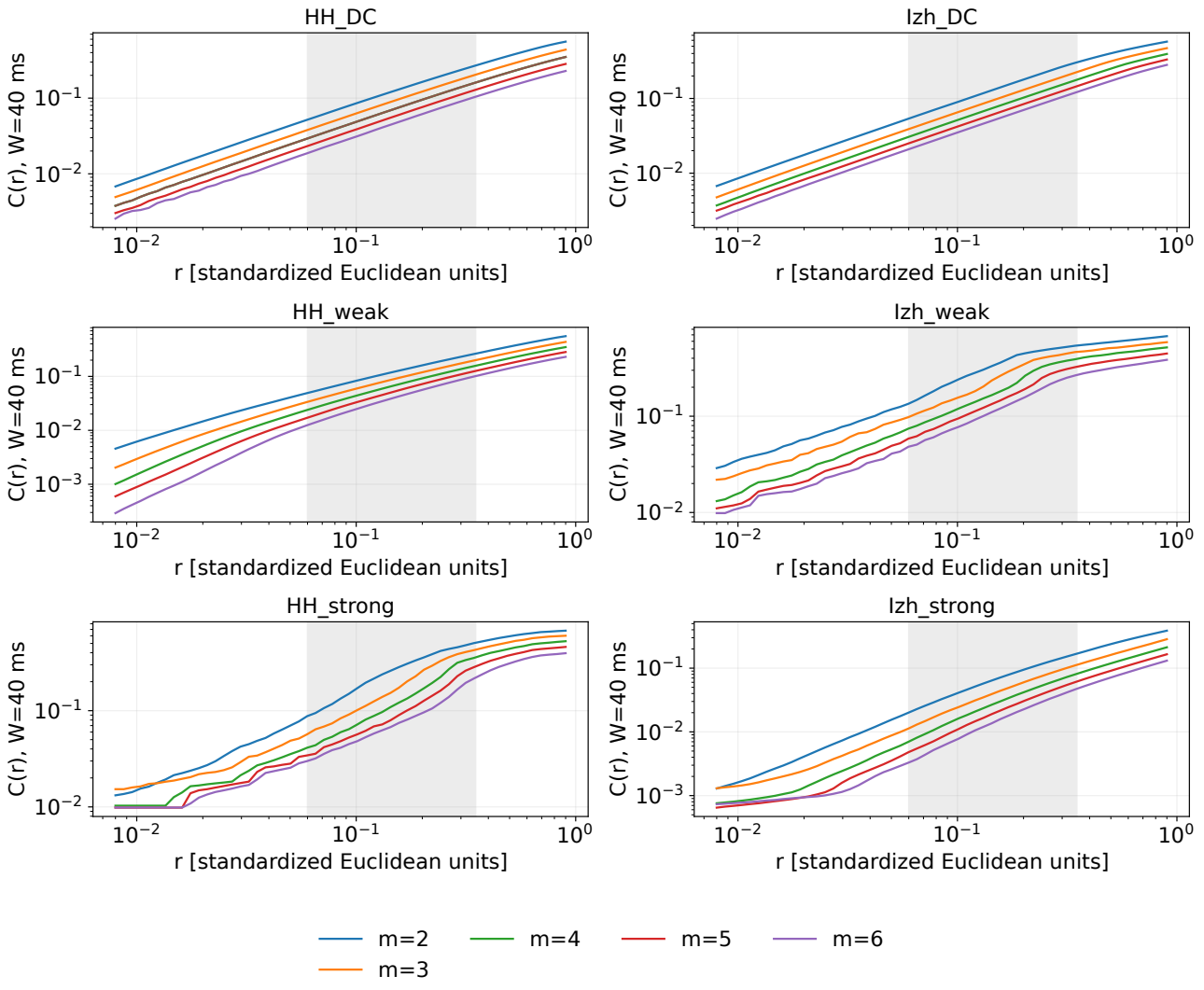


Figure 34: Correlation sums

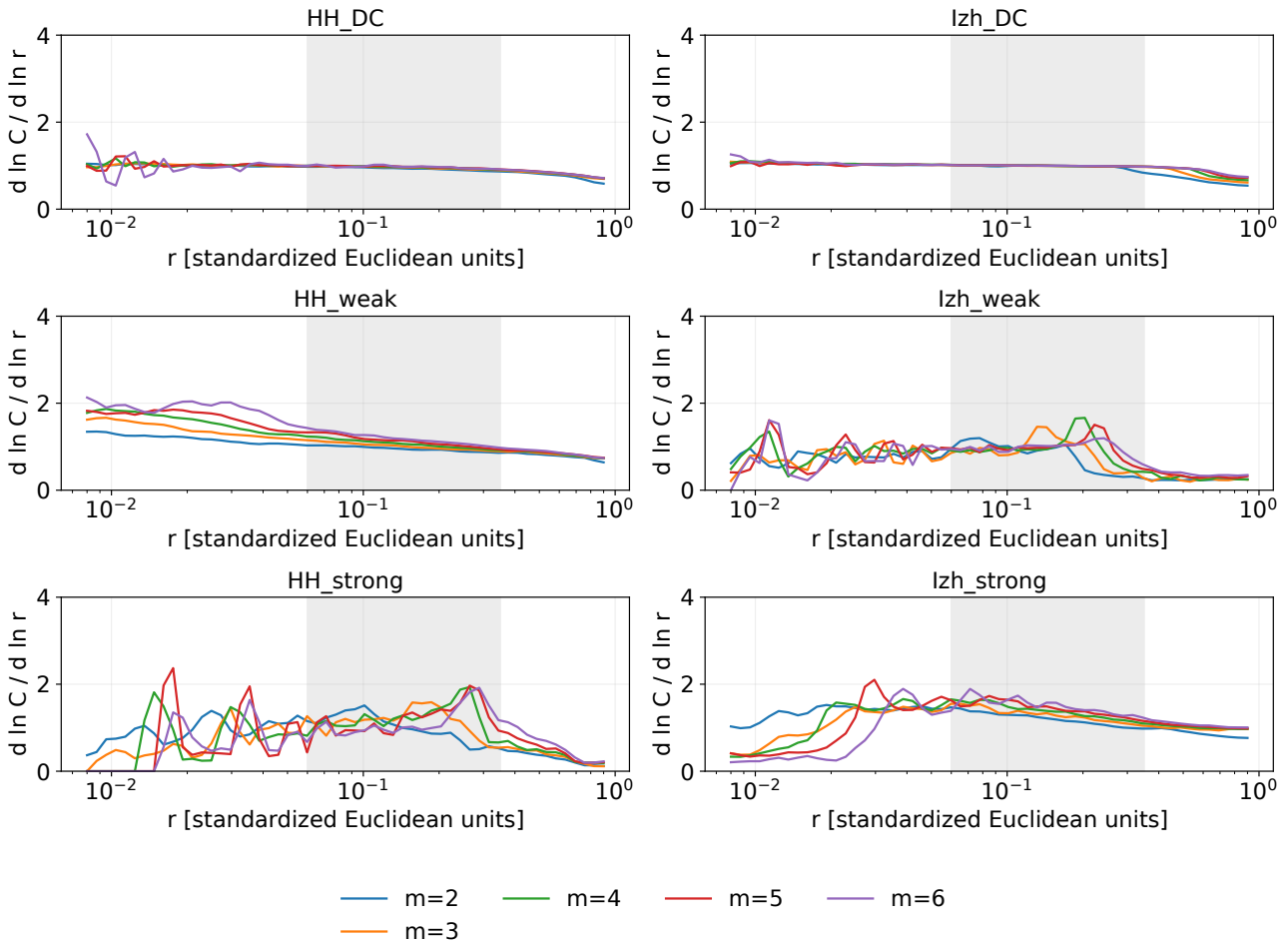


Figure 35: Local correlation slopes

Box-counting uses epsilon=0.025-0.7, fitting 0.08-0.4 with fewer than N/2 occupied boxes. Dimensions m=2,3,4, half/full records and four grid offsets are compared.

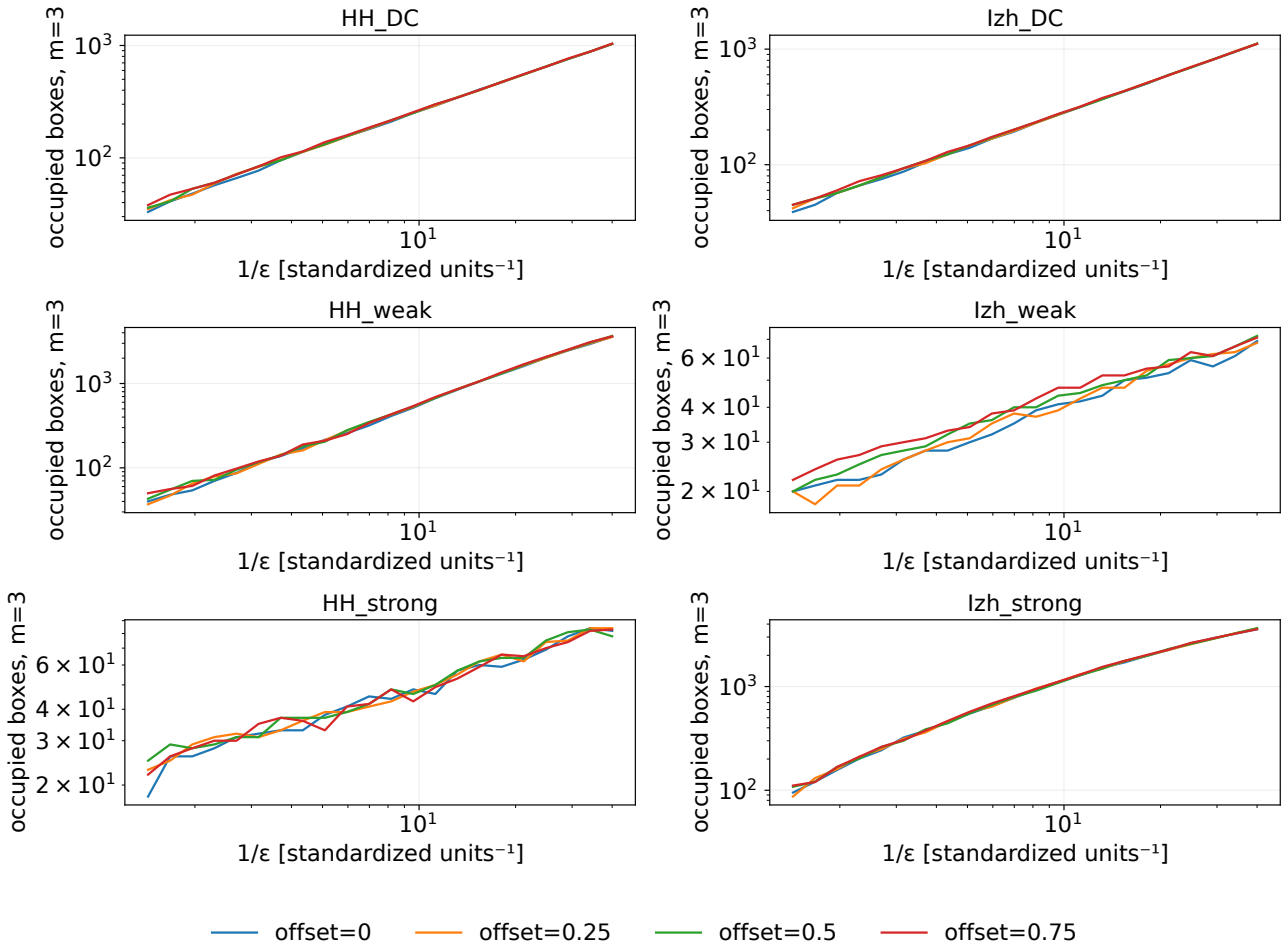


Figure 36: Box-counting curves

signal	rate Hz	ISI CV
HH_DC	68.324	7.2547e-11
Izh_DC	51.513	9.5866e-12
HH_weak	66.327	0.073003
Izh_weak	50	1.1383e-15
HH_strong	50	2.2746e-15
Izh_strong	85.675	0.6008

Raw estimates retain explicit reliability labels:

signal	D2 fit	D2 status	D0 raw m3	D0 status
HH_DC	0.96824	consistent with 1	1.0083	consistent with 1
Izh_DC	0.99823	consistent with 1	0.98196	consistent with 1
HH_weak	1.0678	unreliable: no broad plateau	1.3922	unreliable: parameter sensit ivity
Izh_weak	0.93168	finite-scale; 100 observed phases	0.39257	unreliable: 100 observed phases
HH_strong	1.1502	finite-scale; 100 observed phases	0.32575	unreliable: 100 observed phases
Izh_strong	1.379	unreliable: D2>D0	1.1448	unreliable: D0<D2

signal	D2 m3to6 min	D2 m3to6 max	D2 all settings min	D2 all settings max	D0 all min	D0 all max
HH_DC	0.96539	0.97571	0.95748	0.97571	0.9815	1.0428
HH_strong	0.9986	1.1502	0.97123	1.3109	0.27428	0.34691

signal	D2 m3to6 min	D2 m3to6 max	D2 all settings min	D2 all settings max	D0 all min	D0 all max
HH_weak	1.0273	1.1757	0.98684	1.1757	1.211	1.4946
Izh_DC	0.99736	0.99971	0.99087	0.99971	0.95339	1.0069
Izh_strong	1.288	1.5158	1.222	1.5234	0.93984	1.3408
Izh_weak	0.90637	0.98115	0.82741	1.004	0.27938	0.50403

These are sensitivity ranges, not confidence intervals. A smooth periodic orbit or finite union of reset arcs has dimension 1. A quasiperiodic forced response could have dimension 2, but that requires independent support. Fractional fit digits are not asserted as proven fractal dimensions. The model-based classification of Izh strong does not automatically make its scalar dimension estimate reliable. For the same Izh-strong m=3,1-ms delay and N=40001, GP gives 1.28796 while four box fits give 1.12909-1.16050. They cannot simultaneously be limiting dimensions of the same reconstructed set; the necessary inequality D2<=D0 exposes finite-scale bias.

Phase aliasing and corrected capacity diagnostics

For locked HH strong and Izh weak, T/dt=100. A perfectly periodic record then repeats only 100 observed phases however long it is, producing spurious capacity slopes around 0.33-0.39. These original values are unreliable. New integrations use dt=0.07,h=0.01, observation 7000 ms (2000 forcing phases), and dt=0.013,h=0.001,observation 2600 ms (20000 phases), after 2000 ms transient. Delay remains approximately 1 ms; exact values are retained. All state components are saved. These are explicitly separate sampling checks, not replacements of the common primary observation protocol.

Detailed numerical evidence: Appendix A.6. All columns and original row identifiers are retained.

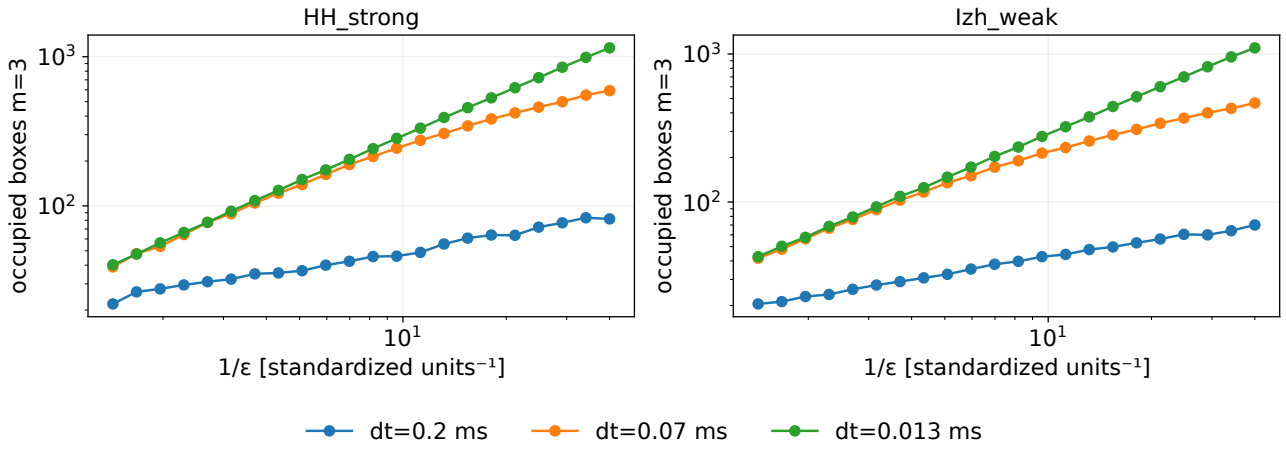


Figure 37: Phase-sampling aliasing and dimension sensitivity

The finest-sampling summaries are:

signal	dt ms	D0 m3 offset mean	D0 min	D0 max
HH_strong	0.013	1.0152	0.99458	1.0321
Izh_weak	0.013	0.98381	0.97536	0.99426

The corrected values support a periodic curve, while grid and scale variation remain visible. They do not retrospectively validate the original aliased fits. At fixed dt=0.2, finer integration alone also checks scalar characteristics:

Detailed numerical evidence: Appendix A.7. All columns and original row identifiers are retained.

Thus integration error and observation aliasing are kept distinct.

Entropy-complexity and control recordings

Voltage patterns use ell=4,5, element delays q=1,5,25, sampling dt=0.2,0.4,1 ms and four consecutive segments. All use the full ell! alphabet, stable temporal index ties and the same Jensen-Shannon normalization and theoretical bounds as C2. Primary comparisons use ell=5,q=5,dt=0.2 (physical delay 1 ms). Matched-length Gaussian, quasiperiodic and shuffled controls are saved with generator states.

signal	H	H sd	C	C sd	patterns per possible
HH_DC	0.33813	0.00070579	0.26976	0.00050051	83.167
HH_strong	0.54942	0	0.37892	0	83.167
HH_weak	0.33032	0.00084547	0.26384	0.00037584	83.167
Izh_DC	0.31349	0.00035435	0.25905	0.00024669	83.167
Izh_strong	0.51303	0.00080138	0.34485	0.00027083	83.167
Izh_weak	0.44481	0	0.35326	0	83.167

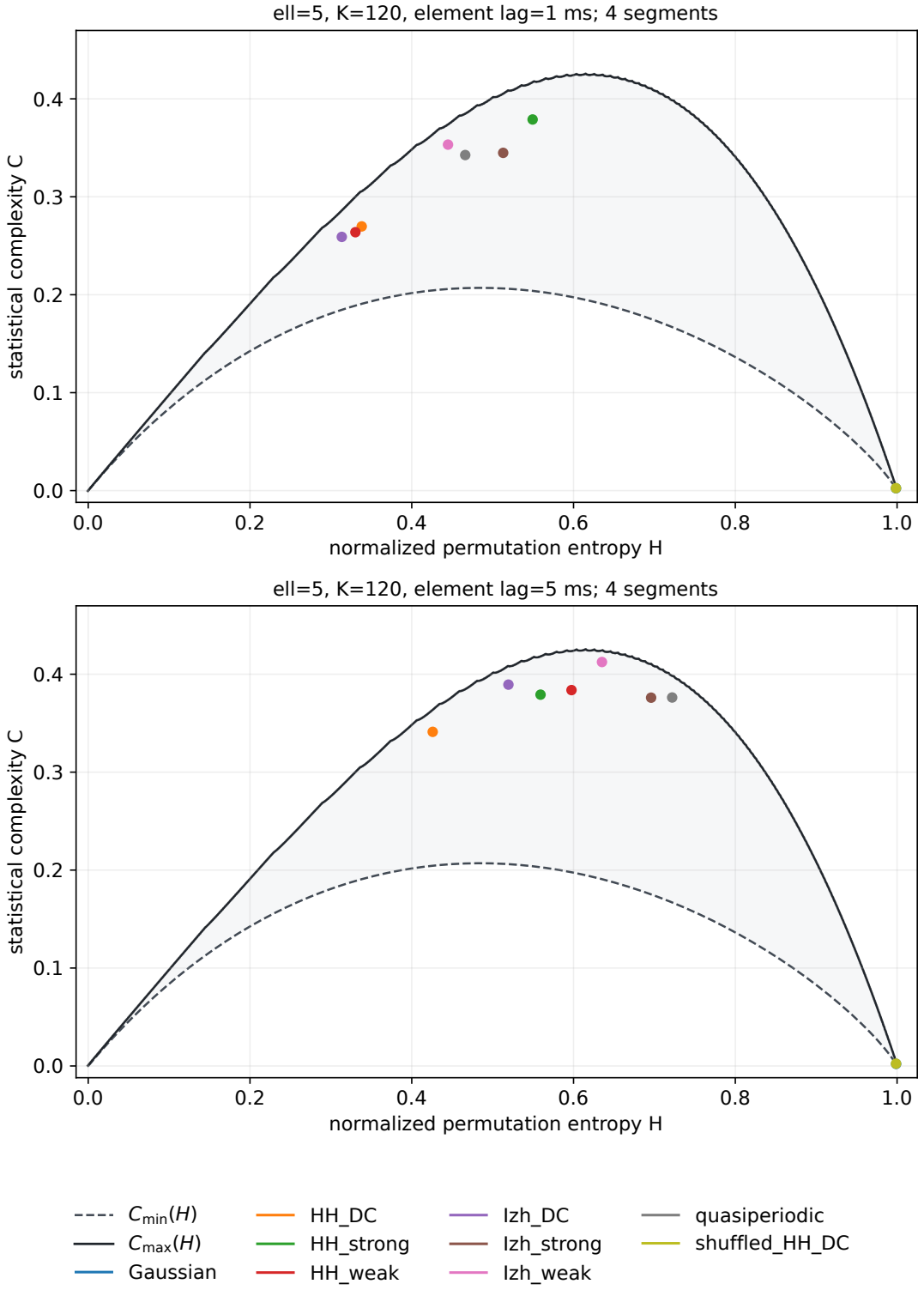


Figure 38: Matched neuronal and control recordings with K=120 theoretical bounds

The main segments have about 83 windows per possible pattern, but dependence and overlap make this a descriptive count, not proof of negligible bias. Different potential waveforms, including the Izh reset, can change ordinal statistics even without different instability. In this set periodic HH strong has higher H at ell=5,q=5 than chaotic Izh strong: entropy is not monotonic in chaoticity.

Ordinal structure of interspike intervals

ISI patterns use ell=3,4,5 and consecutive event intervals, not uniform voltage samples. The table below reports ell=3 at 0.001-ms resolution; the figure compares ell=3 and ell=5. All three lengths and both resolutions are retained in `tables/ISI_ordinal.csv`. Almost periodic ISI order can be dominated by numerical event-time jitter. Both raw intervals and intervals rounded to a stated 0.001-ms resolution are analyzed, with ties handled deterministically. Rounding is not hidden denoising: the raw comparison and an event-accuracy check are retained.

Detailed numerical evidence: Appendix A.8. All columns and original row identifiers are retained.

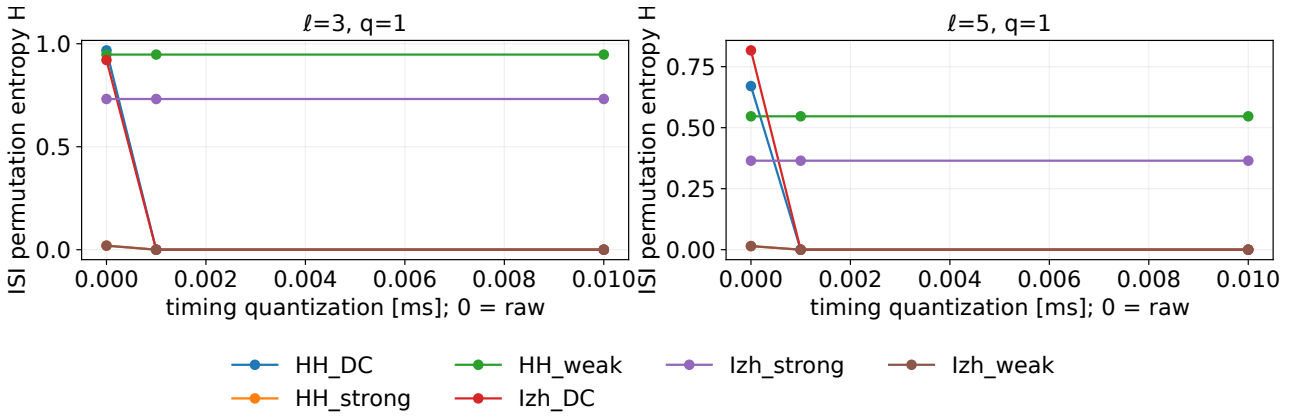


Figure 39: Raw and explicitly rounded ISI ordinal statistics

Values for an almost constant train should not be interpreted as dynamical disorder without inspecting CV, event precision and tie fractions.

5.3.4. What is preserved and what is lost

Detailed numerical evidence: Appendix A.9. All columns and original row identifiers are retained.

The fixed affine calibration substantially improves HH-rate matching relative to identity input on both training and held-out currents. It does not simultaneously preserve low-current silence, sustained-firing threshold and exact f-I rates. Alternative training constraints expose this tradeoff; impossibility for every affine criterion is not proved. Under common periodic inputs, locking regions, spikes per cycle, ISIs and stroboscopic geometry differ. At A=10,T=20, HH is locked with a negative state-tangent exponent, while Izh has robustly positive finite-time estimates. The simplification reproduces some regular firing and entrainment properties of an individual neuron in the tested range, but quantitative threshold, waveform, frequency and synchronization-boundary prediction needs other calibration or a more flexible model. Networks and computational-speed benefits were not tested. Remaining uncertainty includes unresolved periods, finite exponent durations, nonunique calibration criteria, aliasing, discontinuous observables and unreliable dimension plateaus. These limitations are not replaced by an ordinal classifier.

Supplementary reproducibility and references

`models.py` contains RK4, event splitting and independent DOP853 variational maps. `run.py` computes calibration, sweeps, refined orbits and scalar comparisons. `forced_lyapunov.py`, `reliability_checks.py`, `sampling_check.py`, `threshold_checks.py` and `check_calibration.py` produce the additional evidence. Raw states, event times and numerical settings are retained alongside the computed tables. Time is reconstructed as arange(N)*dt_ms after the transient; forcing phase is fixed by an integer number of transient cycles. Versions: python 3.11.4 (v3.11.4:d2340ef257, Jun 6 2023, 19:15:51) [Clang 13.0.0 (clang-1300.0.29.30)], numpy 1.26.4, scipy 1.16.3, numba 0.62.1, matplotlib 3.10.8, pandas 2.2.3. Calibration differential evolution uses legacy MT19937 seed 0; all 624 state words and position are saved, with additional check states in `data/calibration_check_rng_states.json`. Controls use saved PCG64 states. Integration, GP reference selection and FNN are deterministic.

- Hodgkin and Huxley (1952), A quantitative description of membrane current and its application to conduction and excitation in nerve, Journal of Physiology 117:500-544. The shifted voltage convention and exact constants are those of the assignment.
- Izhikevich (2003), Simple model of spiking neurons. The prescribed parameters are retained, with event localization more accurate than the illustrative Euler implementation.
- Kong, Payne, Zhu and Johnson (2024), Saltation Matrices: The Essential Tool for Linearizing Hybrid Dynamical Systems, Proceedings of the IEEE 112:585-608, DOI 10.1109/JPROC.2024.3440211. The applied formula is stated and checked against independent finite differences.
- Reconstruction, FNN, GP, Theiler exclusion and ordinal complexity follow the primary sources listed in the main report and the supplied course definitions.

Appendix A. Detailed numerical tables

These tables provide the full numerical evidence referenced in the main text. Wide tables use portrait panels with repeated identifiers and row numbers.

Appendix A.1: detailed results for C5. Measurement noise, paired Monte Carlo and noisy dimensions

Table, panel 1 of 3. Match rows across panels using the row number.

Row	signal	eta	ell	q	H mean
1	chaotic	0	5	10	0.394139
2	chaotic	0.01	5	10	0.526014
3	chaotic	0.05	5	10	0.797241
4	chaotic	0.1	5	10	0.879682
5	chaotic	0.2	5	10	0.936486
6	spiking	0	5	10	0.343953
7	spiking	0.01	5	10	0.37827
8	spiking	0.05	5	10	0.621485
9	spiking	0.1	5	10	0.753466
10	spiking	0.2	5	10	0.865236

Table, panel 2 of 3. Match rows across panels using the row number.

Row	signal	eta	H sd	C mean	C sd
1	chaotic	0	not estimated	0.301186	not estimated
2	chaotic	0.01	0.000923217	0.304779	0.000345341
3	chaotic	0.05	0.000369452	0.204061	0.000418764
4	chaotic	0.1	0.000586898	0.138154	0.000596896
5	chaotic	0.2	0.000309299	0.0853999	0.000394876
6	spiking	0	not estimated	0.277995	not estimated
7	spiking	0.01	0.00058473	0.290674	0.000186382
8	spiking	0.05	0.000796239	0.313246	0.000501699
9	spiking	0.1	0.000893377	0.260946	0.000564696
10	spiking	0.2	0.00069543	0.175204	0.000763635

Table, panel 3 of 3. Match rows across panels using the row number.

Row	signal	eta	realizations
1	chaotic	0	1
2	chaotic	0.01	5
3	chaotic	0.05	5
4	chaotic	0.1	5
5	chaotic	0.2	5
6	spiking	0	1
7	spiking	0.01	5
8	spiking	0.05	5
9	spiking	0.1	5
10	spiking	0.2	5

Appendix A.2: detailed results for D. Overall synthesis and critical discussion

Table, panel 1 of 3. Match rows across panels using the row number.

Row	origin	I	classification	lambda max
1	HR spiking	4	regular spiking	3.438e-05
2	HR bursting	2.5	periodic bursting	4.880e-06

Row	origin	I	classification	lambda max
3	HR chaotic	3.25	deterministic chaos: numerical evidence	0.0135346
4	HR steady	1	stable equilibrium	-0.012149
5	white_noise	not applicable	IID Gaussian	not applicable
6	shuffled	not applicable	random permutation of HR chaos	not applicable
7	quasiperiodic	not applicable	two incommensurate frequencies	not applicable
8	HR spiking + measurement noise eta=.05	4	periodic source + measurement noise	source model approximately 0

Table, panel 2 of 3. Match rows across panels using the row number.

Row	origin	I	k; m; W	D2	D0
1	HR spiking	4	37; 3; 148	0.970-1.004: consistent with 1	1.017-1.050: finite-scale
2	HR bursting	2.5	125; 5; 500	0.980-0.989: consistent with 1	not estimated
3	HR chaotic	3.25	10; 6; 656	1.819-1.928: finite-scale; IC2 1.910	1.874-1.941: polyline, finite-scale; points 1.689-1.747 unreliable
4	HR steady	1	not applicable	0: theory	0: theory
5	white_noise	not applicable	not estimated	not estimated	not estimated
6	shuffled	not applicable	not estimated	not estimated	not estimated
7	quasiperiodic	not applicable	not estimated	not estimated	not estimated
8	HR spiking + measurement noise eta=.05	4	37; 2-6; 148	no common plateau; see C5	not estimated

Table, panel 3 of 3. Match rows across panels using the row number.

Row	origin	I	H	C
1	HR spiking	4	0.343953	0.277995
2	HR bursting	2.5	0.353274	0.279879
3	HR chaotic	3.25	0.394139	0.301186
4	HR steady	1	not applicable	not applicable
5	white_noise	not applicable	0.999937	0.000113417
6	shuffled	not applicable	0.999938	0.000111083
7	quasiperiodic	not applicable	0.556953	0.381531
8	HR spiking + measurement noise eta=.05	4	0.621485	0.313246

Appendix A.3: detailed results for Operational thresholds and transient false positives

Table, panel 1 of 5. Match rows across panels using the row number.

Row	model	old definition	J uA cm2	transient ms	duration ms
1	HH	short	6.2602	1500	3000
2	HH	long	6.2603	3000	6000

Table, panel 2 of 5. Match rows across panels using the row number.

Row	model	old definition	data file	old total count passed	sustained
1	HH	short	data/old_threshold_false_positive_HH_short.npz	True	False
2	HH	long	data/old_threshold_false_positive_HH_long.npz	True	False

Table, panel 3 of 5. Match rows across panels using the row number.

Row	model	old definition	total events	window count min	window rate relative range
1	HH	short	7	0	—
2	HH	long	15	0	—

Table, panel 4 of 5. Match rows across panels using the row number.

Row	model	old definition	tail median ISI ms	last gap ms	last event ms
1	HH	short	20.169	2867.3	132.69
2	HH	long	20.118	5715.6	284.35

Table, panel 5 of 5. Match rows across panels using the row number.

Row	model	old definition	last gap in tail ISI
1	HH	short	142.17
2	HH	long	284.1

Appendix A.4: detailed results for 5.3.2. Periodic forcing, synchronization and transitions

model	A min	A max	period cycles	mean spikes per cycle	grid points
HH	0	1.1	0	1.3496	12
HH	1.2	1.2	7	0.995	1
HH	1.3	1.6	0	0.995	4
HH	1.7	2.7	2	1	11
HH	2.8	15	1	1	123
Izh	0	0.3	0	1.025	4
Izh	0.4	2.8	1	1	25
Izh	2.9	2.9	0	1.11	1
Izh	3	3	7	1.14	1
Izh	3.1	3.1	0	1.155	1
Izh	3.2	3.3	6	1.1675	2
Izh	3.4	3.4	0	1.19	1

model	A min	A max	period cycles	mean spikes per cycle	grid points
Izh	3.5	3.7	5	1.2	3
Izh	3.8	3.9	0	1.23	2
Izh	4	4.3	4	1.25	4
Izh	4.4	4.4	8	1.25	1
Izh	4.5	4.6	7	1.285	2
Izh	4.7	4.8	0	1.31	2
Izh	4.9	5.6	3	1.335	8
Izh	5.7	5.7	8	1.375	1
Izh	5.8	5.8	0	1.385	1
Izh	5.9	6.1	5	1.4	3
Izh	6.2	6.2	0	1.42	1
Izh	6.3	6.3	7	1.425	1
Izh	6.4	6.7	0	1.4625	4
Izh	6.8	8.2	2	1.5	15
Izh	8.3	8.3	0	1.555	1
Izh	8.4	8.4	7	1.57	1
Izh	8.5	8.5	0	1.585	1
Izh	8.6	8.8	5	1.6	3
Izh	8.9	8.9	8	1.625	1
Izh	9	9.1	0	1.645	2
Izh	9.2	9.9	3	1.665	8
Izh	10	10	0	1.71	1
Izh	10.1	10.1	7	1.715	1
Izh	10.2	10.2	0	1.73	1
Izh	10.3	10.7	4	1.75	5
Izh	10.8	10.8	0	1.78	1
Izh	10.9	11.2	5	1.8	4
Izh	11.3	11.5	6	1.8333	3
Izh	11.6	11.7	7	1.86	2
Izh	11.8	11.8	8	1.875	1
Izh	11.9	12.4	0	1.9275	6
Izh	12.5	15	1	2	26

Appendix A.5: detailed results for Refined periodic solutions and hybrid stability

Table, panel 1 of 3. Match rows across panels using the row number.

Row	model	T ms	A uA cm2	root success	residual max
1	HH	20	10.1	True	1.08e-12
2	Izh	20	13.8	True	1.4211e-14

Table, panel 2 of 3. Match rows across panels using the row number.

Row	model	T ms	spectral radius	multipliers	FD relative error
1	HH	20	0.025365	[-1.15670012e-02 2.53654208e-02 4.28436153e-18 -6.18253916e-15]	2.8267e-06
2	Izh	20	0.70375	[-8.99280650e-15 7.03749021e-01]	4.3916e-10

Table, panel 3 of 3. Match rows across panels using the row number.

Row	model	T ms	tight tolerance matrix difference	wrong DR only spectral radius	events per period
1	HH	20	1.6727e-10	—	—
2	Izh	20	1.0837e-11	0.5816	2

Appendix A.6: detailed results for Phase aliasing and corrected capacity diagnostics

Table, panel 1 of 3. Match rows across panels using the row number.

Row	signal	dt ms	N	actual lag ms	D0 m3 offset mean
1	HH_strong	0.2	40001	1	0.32575
2	HH_strong	0.07	100001	0.98	0.89967
3	HH_strong	0.013	200001	1.001	1.0152
4	Izh_weak	0.2	40001	1	0.39257
5	Izh_weak	0.07	100001	0.98	0.78547
6	Izh_weak	0.013	200001	1.001	0.98381

Table, panel 2 of 3. Match rows across panels using the row number.

Row	signal	dt ms	D0 min	D0 max	D2 m4
1	HH_strong	0.2	0.29935	0.33745	1.1502
2	HH_strong	0.07	0.88587	0.91116	0.95716
3	HH_strong	0.013	0.99458	1.0321	1.1877
4	Izh_weak	0.2	0.37189	0.42009	0.93168
5	Izh_weak	0.07	0.7662	0.80935	0.8925
6	Izh_weak	0.013	0.97536	0.99426	0.94365

Table, panel 3 of 3. Match rows across panels using the row number.

Row	signal	dt ms	max boxes smallest eps	distinct forcing phases theoretical
1	HH_strong	0.2	84	100
2	HH_strong	0.07	600	2000
3	HH_strong	0.013	1153	20000
4	Izh_weak	0.2	72	100
5	Izh_weak	0.07	469	2000
6	Izh_weak	0.013	1108	20000

Appendix A.7: detailed results for Phase aliasing and corrected capacity diagnostics

Table, panel 1 of 6. Match rows across panels using the row number.

Row	signal	coarse h ms	fine h ms	transient ms	observation ms
1	HH_weak	0.01	0.005	2000	8000
2	Izh_strong	0.01	0.005	2000	8000

Table, panel 2 of 6. Match rows across panels using the row number.

Row	signal	coarse h ms	data file	coarse events	fine events
1	HH_weak	0.01	data/reliability_scalar_HH_A1.00_h0.005.npz	531	531
2	Izh_strong	0.01	data/reliability_scalar_Izh_A10.00_h0.005.npz	685	685

Table, panel 3 of 6. Match rows across panels using the row number.

Row	signal	coarse h ms	coarse rate Hz	fine rate Hz	coarse ISI CV
1	HH_weak	0.01	66.327	66.327	0.073003
2	Izh_strong	0.01	85.675	85.684	0.6008

Table, panel 4 of 6. Match rows across panels using the row number.

Row	signal	coarse h ms	fine ISI CV	coarse H	fine H
1	HH_weak	0.01	0.073003	0.33044	0.33044
2	Izh_strong	0.01	0.60033	0.51265	0.5126

Table, panel 5 of 6. Match rows across panels using the row number.

Row	signal	coarse h ms	coarse C	fine C	coarse D2
1	HH_weak	0.01	0.26391	0.26391	1.0678
2	Izh_strong	0.01	0.34461	0.34463	1.379

Table, panel 6 of 6. Match rows across panels using the row number.

Row	signal	coarse h ms	fine D2
1	HH_weak	0.01	1.0678
2	Izh_strong	0.01	1.3798

Appendix A.8: detailed results for Ordinal structure of interspike intervals

Table, panel 1 of 3. Match rows across panels using the row number.

Row	signal	timing rounding ms	ell	ISI count	H
1	HH_DC	0.001	3	546	0
2	Izh_DC	0.001	3	411	0
3	HH_weak	0.001	3	530	0.94746
4	Izh_weak	0.001	3	399	0
5	HH_strong	0.001	3	399	0
6	Izh_strong	0.001	3	684	0.7317

Table, panel 2 of 3. Match rows across panels using the row number.

Row	signal	timing rounding ms	C	patterns	occupied
1	HH_DC	0.001	0	544	1
2	Izh_DC	0.001	0	409	1
3	HH_weak	0.001	0.048761	528	6
4	Izh_weak	0.001	0	397	1
5	HH_strong	0.001	0	397	1
6	Izh_strong	0.001	0.23679	682	4

Table, panel 3 of 3. Match rows across panels using the row number.

Row	signal	timing rounding ms	ties	patterns per possible
1	HH_DC	0.001	1	90.667
2	Izh_DC	0.001	1	68.167
3	HH_weak	0.001	0	88
4	Izh_weak	0.001	1	66.167
5	HH_strong	0.001	1	66.167

Row	signal	timing rounding ms	ties	patterns per possible
6	Izh_strong	0.001	0	113.67

Appendix A.9: detailed results for 5.3.4. What is preserved and what is lost

Table, panel 1 of 3. Match rows across panels using the row number.

Row	signal	classification	lambda max per ms	D2 fit
1	HH_DC	autonomous regular firing; neutral time-shift direction	-0.00010632	0.96824
2	Izh_DC	autonomous regular firing; neutral time-shift direction	-5.9565e-05	0.99823
3	HH_weak	period <=8 unresolved; near-neutral exponent; quasiperiodicity plausible, unproved	-4.1601e-05	1.0678
4	Izh_weak	stable stroboscopic period 1T; λ<0	-0.01553	0.93168
5	HH_strong	stable stroboscopic period 1T; λ<0	-0.18367	1.1502
6	Izh_strong	numerically chaotic: positive model exponent at three steps and in every 8-s block	0.0018698	1.379

Table, panel 2 of 3. Match rows across panels using the row number.

Row	signal	D2 status	D0 raw m3	D0 status
1	HH_DC	consistent with 1	1.0083	consistent with 1
2	Izh_DC	consistent with 1	0.98196	consistent with 1
3	HH_weak	unreliable: no broad plateau	1.3922	unreliable: parameter sensitivity
4	Izh_weak	finite-scale; 100 observed phases	0.39257	unreliable: 100 observed phases
5	HH_strong	finite-scale; 100 observed phases	0.32575	unreliable: 100 observed phases
6	Izh_strong	unreliable: D2>D0	1.1448	unreliable: D0<D2

Table, panel 3 of 3. Match rows across panels using the row number.

Row	signal	D0 phase rich m3	H ell5 q5	C ell5 q5
1	HH_DC	—	0.33812	0.26975
2	Izh_DC	—	0.31361	0.25914
3	HH_weak	—	0.33044	0.26391
4	Izh_weak	0.98381	0.44454	0.35308
5	HH_strong	1.0152	0.54938	0.3789
6	Izh_strong	—	0.51265	0.34461